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Toolpath Fingerprinting from CNC Sensor Data Using Air-Cut Referencing: A Temporal Binning Approach

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Abstract: We propose air-cut referencing, a methodology for isolating toolpath-specific sensor signatures from CNC machining data by normalizing active-cutting measurements against air-cut (no-load) baselines. Using temporal binning to align program phases, we extract statistical features and evaluate four signal isolation methods—raw, mean-subtracted, ratio-normalized, and z-score—for three-class toolpath classification (adaptive clearing, facing, pocketing) with six Arduino Nano 33 BLE Sense Lite IMU modules (LSM9DS1) and four Hall-effect current sensors. Physical interpretation reveals that the dominant discriminative features are magnetometer channels encoding workspace position rather than cutting dynamics. Critically, when all position-confounded channels are excluded (784 features), raw features degrade to 96.7% (minimum fold 81.8% over 100 repeated CV evaluations) while the subtracted and ratio methods maintain perfect 100% accuracy and z-score remains near-perfect (99.0%, minimum 90%)—a smaller but genuine, confound-independent benefit. On a position-controlled damaged-tool detection task (the same G-code programs run with a damaged tool; 51 normal vs. 14 damaged runs), accuracy reaches 98.5% (balanced accuracy 0.99) and the discriminative signal shifts decisively from the magnetometer (0.1% of importance) to the accelerometer and acoustic-RMS channels (74% combined), confirming that the sensors capture genuine process dynamics once workspace position is held constant. Cross-condition transfer (37% accuracy, near chance) confirms that air and active conditions produce fundamentally different feature distributions. Comprehensive ablation shows a single bed-mounted IMU suffices, 5 training runs per class achieve 96%, and referenced features resist additive noise up to 1.0× the feature standard deviation. These results establish air-cut referencing as a robust normalization strategy for CNC sensor data, while the physical interpretation analysis provides a template for distinguishing position-encoding confounds from process-relevant sensor signals.

Keywords: CNC machining; toolpath classification; air-cut referencing; sensor fusion; signal isolation; temporal binning

1. Introduction

Computer numerical control (CNC) machines produce sensor signals that mix two contributions: machine-specific characteristics (motor vibrations, structural resonances, drive electronics) and process-specific characteristics (cutting forces, tool engagement, material removal). Models trained to recognize machining operations from sensor data typically learn both, which makes them fragile—they fail when deployed on a different machine [1,2]. As production environments become more connected through Industry 4.0 initiatives [3,4], this machine-dependence problem limits the practical utility of learned models for quality control, predictive maintenance, and manufacturing security applications, all of which would benefit from sensor-based toolpath verification that transfers across equipment.

This paper proposes *air-cut referencing*: run the same toolpath program twice, once with material (active cut) and once without (air cut), then use the air-cut measurements as a baseline to normalize

away the machine-specific component, leaving the process-specific signature. The methodology is evaluated on three toolpath types (adaptive clearing, facing, pocketing) using 106 sensor recordings from a Bantam Tools desktop CNC instrumented with six Arduino Nano 33 BLE Sense Lite IMU modules (LSM9DS1 9-axis sensors plus LPS22HB pressure/temperature, APDS-9960 color/proximity, and an onboard PDM microphone) and four Hall-effect motor current sensors. Because the air and active runs in our dataset used different CAM-generated G-code programs, point-by-point subtraction is impossible, so we introduce a *temporal binning* scheme that normalizes each run's time axis to $[0, 1]$, divides it into 20 bins, and computes seven summary statistics per sensor per bin. Four isolation methods are compared: raw features (control), subtracted (active – air mean), ratio (active / air mean), and z-score $((\text{active} - \mu_{\text{air}}) / \sigma_{\text{air}})$.

A trivial baseline analysis upfront shows the three-class task is inherently easy: G-code line range alone achieves 100% accuracy and mean axis position achieves 98%, because the three CAM-generated programs traverse different regions of the machine's workspace. We therefore do not claim the headline 100% Random Forest accuracy as the paper's contribution. Instead, we dissect what the sensors are actually detecting and find that the top-ranked features are magnetometer channels acting as *workspace-position encoders*—they pick up the machine's static magnetic field as the sensor moves through different regions, rather than detecting cutting dynamics. At the 4 Hz effective sampling rate, the system cannot capture cutting vibration anyway (tooth-passing frequency is 400 Hz, far above the 2 Hz Nyquist limit), so the accelerometer features encode trajectory dynamics and gravitational tilt, not cutting forces. We explicitly distinguish “trajectory-relevant” from “force-relevant” interpretations and connect this to the UHMW-PE workpiece (cutting forces only $\sim 1\text{--}2\text{ N}$), the LSM9DS1 bias floor (10–50 mg, comparable to the observed $\sim 20\text{ mg}$ inter-class differences), and the desktop CNC's stepper-driven architecture which differs fundamentally from industrial servo-driven machines.

Despite the easy task and the magnetometer confound, air-cut referencing turns out to provide *genuine, confound-independent benefit*. The decisive experiment removes ALL position-confounded channels (magnetometers, color, pressure, proximity) and re-runs the full 20×5 -fold repeated cross-validation pipeline on the remaining 784 features (accelerometer, gyroscope, RMS, temperature, electrical current). Raw features drop to 96.7% mean accuracy with a minimum fold of 81.8%, while the subtracted and ratio methods maintain **perfect 100% accuracy across all 100 evaluations** and z-score remains near-perfect (99.0%, minimum 90%). The effect is larger for the noise-sensitive MLP—raw MLP reaches only 79.9% while z-score MLP recovers to 90.0%. A 1,000-permutation test rules out chance performance (null max 62.7% vs. observed 100%, $p < 0.001$), Friedman testing confirms statistical significance ($\chi^2 = 33.0$, $p < 10^{-6}$, Bonferroni-corrected pairwise $p = 0.011$ for raw vs. each referenced method), and noise robustness analysis shows referenced features maintain accuracy up to $1.0 \times$ additive noise while raw features degrade. A single bed-mounted IMU achieves 100% accuracy with only 238 features, just 5 training runs per class suffice for 96.4%, and an error analysis traces the one stubbornly-misclassified run to a color-sensor anomaly likely caused by ambient lighting changes—exactly the kind of session-specific noise that air-cut referencing is designed to normalize. Finally, on a *position-controlled* damaged-tool detection task—the same G-code programs run with a damaged tool, so workspace position is held constant—Random Forest reaches 98.5% accuracy (balanced 0.99) and the discriminative signal shifts from the magnetometer (0.1% of importance) to the accelerometer and acoustic-RMS channels (74% combined), demonstrating that the sensors capture genuine process dynamics once the position confound is removed. Validating transfer across machines remains future work.

The contributions of this paper are therefore methodological and practical rather than performance-driven:

1. We introduce an *air-cut referencing methodology* with temporal binning that isolates process-relevant signal components from machine-specific and workspace-position-dependent baselines, even when air-cut and active-cutting data originate from different CAM-generated G-code programs.

2. We provide *rigorous statistical evidence* that air-cut referencing offers benefits beyond raw feature classification, demonstrated through repeated cross-validation, permutation testing, Friedman/Wilcoxon analysis with Bonferroni correction, bootstrap confidence intervals, Cliff's delta effect sizes, and noise robustness curves.
3. We perform a *transparent physical interpretation* that distinguishes position-encoding confounds from process-relevant features, including the demonstration that air-cut referencing retains its classification benefit (subtracted and ratio at 100%, z-score at 99.0%, vs. raw 96.7%) even after all position-confounded channels are excluded from the feature set.
4. We present a *comprehensive ablation and characterization study*, including sensor subset analysis, temporal bin importance, learning curves, cross-condition transfer, leave-one-type-out novelty detection, comparison against ROCKET-style time-series methods, and per-channel temporal drift analysis. Together these establish a template for honestly evaluating sensor-based classification systems and a set of practical sensor deployment guidelines.

The remainder of this paper is organized as follows. Section 2 reviews related work in CNC process monitoring, sensor fusion, and machine learning for manufacturing. Section 3 details the air-cut referencing methodology, temporal binning, and classification framework. Section 4 describes the experimental apparatus and data collection. Section 5 presents classification results and physical interpretation, Section 6 reports the ablation study, Section 7 discusses implications and limitations, and Section 8 concludes.

2. Related Work

2.1. Sensor-Based Process Monitoring in CNC Machining

Sensor-based monitoring of CNC machining processes has been an active research area for over three decades. Teti et al. [3] provided a comprehensive review covering force, vibration, acoustic emission, and current sensors for machining operations. Byrne et al. [4] established foundational work on tool condition monitoring using multiple sensor modalities. More recently, Nath [5] reviewed integrated sensor systems for advanced manufacturing, and Kuntoğlu et al. [6] surveyed sensor-based approaches specifically for Industry 4.0 CNC monitoring. The integration of low-cost MEMS-based inertial measurement units (IMUs) has expanded the range of deployable sensor configurations [7], enabling dense instrumentation of machine structures at modest cost.

A recurring finding is that sensor signals during machining are influenced by both process parameters (feed rate, depth of cut, tool engagement) and machine-specific factors (structural dynamics, bearing condition, drive characteristics). While most monitoring approaches treat the combined signal as the object of interest, our work explicitly decomposes these contributions through air-cut referencing.

2.2. Sensor Fusion for Manufacturing

Multi-sensor data fusion has demonstrated significant advantages over single-sensor approaches for machining process characterization [8,9]. Combining heterogeneous sensor types—such as accelerometers with current sensors—captures complementary aspects of the process. Feature-level fusion, where statistical features from individual sensor channels are concatenated into a joint feature vector, is the most common strategy and is the approach adopted in this work. Zhou et al. [10] reviewed multi-sensor fusion architectures for intelligent manufacturing, while deep-learning-based fusion has shown promise for larger datasets [11].

2.3. Signal Processing and Feature Extraction

Time-domain statistical features (mean, variance, RMS, kurtosis, skewness) and frequency-domain features (spectral energy, dominant frequency, power spectral density) are widely used for characterizing machining signals [12–14]. Rmili et al. [15] demonstrated the effectiveness of statistical

and spectral features for automatic tool wear classification. Our temporal binning approach is related to the concept of short-time feature extraction, where signals are divided into temporal segments and statistics are computed per segment.

2.4. Machine Learning for Toolpath and Process Classification

Random Forests [16], Support Vector Machines [17,18], and neural networks have all been applied to machining process classification. Bustillo et al. [19] demonstrated smart manufacturing applications of ensemble methods. Wang et al. [20] and Cao et al. [21] reviewed deep learning approaches for manufacturing quality prediction. Bergs et al. [22] positioned machine learning within the broader digital twin framework for manufacturing.

Recent work on time-series classification offers alternative approaches to the bag-of-statistics paradigm used here. Dempster et al. [23] introduced ROCKET (Random Convolutional Kernel Transform), which applies random convolutional filters directly to time series and achieves competitive accuracy with minimal computational cost. Fawaz et al. [24] proposed InceptionTime, a deep learning architecture for time-series classification. We compare our temporal binning approach against ROCKET-style features in Section 5.

2.5. Transfer Learning and Domain Adaptation

The cross-machine generalization problem—where models trained on one machine fail to transfer to another—has been addressed through transfer learning [2,25] and domain adaptation [26]. These approaches typically require some labeled data from the target domain. Our air-cut referencing methodology offers a complementary approach: by normalizing against a machine-specific baseline, the resulting features may be inherently more transferable, though this hypothesis requires cross-machine validation (Section 7).

2.6. Air-Cut and No-Load Testing

Air cutting (also called dry running or no-load testing) is standard practice in machine tool acceptance testing and calibration [27]. It characterizes spindle dynamics, axis positioning accuracy, and thermal behavior without cutting forces. Altintas [28] provides comprehensive treatment of machine dynamics characterization through no-load testing. In tool-condition monitoring, subtracting a no-load (idle) spindle- or axis-motor current/power baseline to isolate the cutting-load component is itself a well-established practice [3,4,12]; this is also why we frame our near-constant spindle-current result (Section 5.7) as a known limitation of current-based monitoring under shallow cuts rather than a novel finding. Air-cut referencing generalizes the no-load-baseline idea in two ways: from a single current channel to the full multi-modal, temporally-binned feature vector (110 channels $\times$ per-phase statistics), and from load estimation to the removal of machine- and *workspace-position*-specific offsets prior to classification. To the best of our knowledge, no prior work has systematically used air-cut data as a *reference baseline across the full sensor array* for isolating process-specific signal components to enable toolpath or tool-condition classification.

2.7. CNC Security and Manufacturing Integrity

CNC manufacturing security has gained attention as machines become networked [29]. Sturm et al. [30] and Zeltmann et al. [31] demonstrated that malicious modifications to G-code or process parameters can introduce defects undetectable by conventional inspection. Yampolskiy et al. [32] surveyed security threats across manufacturing modalities. Belikovetsky et al. [33] showed that sensor-based verification can detect sabotaged manufacturing processes. Our toolpath fingerprinting methodology directly supports this application by providing sensor-based toolpath verification.

3. Methodology

This section presents the air-cut referencing methodology for extracting toolpath fingerprints from CNC sensor data. We describe the data acquisition pipeline, temporal binning strategy, signal isolation methods, dimensionality reduction, and classification framework.

3.1. Data Acquisition and Sensor Configuration

We instrument a desktop CNC milling machine with two categories of sensors: six wireless IMU (inertial measurement unit) modules mounted at distinct locations on the machine frame and moving components, and four electrical sensors measuring spindle and axis motor currents. Each IMU module records 17 channels comprising three-axis linear acceleration, three-axis angular velocity, three-axis magnetometer readings, and derived orientation quantities. The electrical sensors capture spindle motor current and current draw plus acceleration for the X, Y, and Z linear axes. Together, these provide 110 features per timestep across all sensor modules.

Data is collected at an approximately 0.25 s sampling interval, synchronized across all sensor modules via timestamp alignment. Each experimental run produces a single CSV file containing the time-aligned multi-sensor data stream for the full duration of the toolpath execution.

We collect data under two conditions:

- **Air-cut runs:** The CNC machine executes the toolpath program with the spindle running but with no workpiece mounted, so the tool moves through air without engaging any material.
- **Active-cutting runs:** The machine executes the toolpath on a TIVAR UHMW-PE (ultra-high molecular weight polyethylene) workpiece, producing actual material removal.

Three toolpath types are studied: *adaptive clearing*, *facing*, and *pocketing*. These represent fundamentally different engagement geometries and cutting force profiles, providing a meaningful classification challenge.

3.2. Temporal Binning and Feature Extraction

3.2.1. Motivation: Why G-Code Alignment Fails

An ideal approach to air-cut referencing would align the air-cut and active-cutting signals on a per-G-code-command basis, enabling direct subtraction at each point in the toolpath. However, this requires that both datasets be generated from the same G-code program with identical command sequences. In our experimental setting, the air-cut and active-cutting programs were generated by different CAM configurations, resulting in different G-code command sequences for the same nominal toolpath type. Feed rates, plunge strategies, and retract moves differ between programs, making point-by-point temporal alignment infeasible.

3.2.2. Temporal Binning Strategy

We address this misalignment through temporal binning, which captures the meso-scale temporal structure of each run without requiring command-level synchronization. Given a sensor data matrix $\mathbf{X} \in \mathbb{R}^{T \times C}$ for a run of T timesteps and C channels, we divide the time axis into B equal-duration bins:

$$\mathbf{X}^{(b)} = \left\{ \mathbf{x}_t \in \mathbb{R}^C \mid t \in \left[\frac{(b-1) \cdot T}{B}, \frac{b \cdot T}{B} \right) \right\}, \quad b = 1, \dots, B \quad (1)$$

where $\mathbf{X}^{(b)}$ denotes the set of timesteps falling within bin b .

3.2.3. Statistical Feature Computation

For each bin b and each channel c , we compute seven summary statistics:

$$\mathbf{f}^{(b,c)} = \begin{bmatrix} \mu_{b,c} \\ \sigma_{b,c} \\ \min_{b,c} \\ \max_{b,c} \\ \text{median}_{b,c} \\ \text{skew}_{b,c} \\ \text{kurt}_{b,c} \end{bmatrix} = \begin{bmatrix} \frac{1}{|\mathbf{X}^{(b)}|} \sum_{t \in b} x_{t,c} \\ \sqrt{\frac{1}{|\mathbf{X}^{(b)}|-1} \sum_{t \in b} (x_{t,c} - \mu_{b,c})^2} \\ \min_{t \in b} x_{t,c} \\ \max_{t \in b} x_{t,c} \\ \text{median}(\{x_{t,c}\}_{t \in b}) \\ \frac{1}{|\mathbf{X}^{(b)}|} \sum_{t \in b} \left(\frac{x_{t,c} - \mu_{b,c}}{\sigma_{b,c}} \right)^3 \\ \frac{1}{|\mathbf{X}^{(b)}|} \sum_{t \in b} \left(\frac{x_{t,c} - \mu_{b,c}}{\sigma_{b,c}} \right)^4 - 3 \end{bmatrix} \quad (2)$$

where $|\mathbf{X}^{(b)}|$ is the number of timesteps in bin b .

3.2.4. Run-Level Feature Vector Assembly

Each of the $7C$ per-bin statistics forms a length- B trajectory across the temporal bins. To obtain a fixed-length run-level descriptor whose dimensionality does not grow with the bin count, we summarize each per-bin statistic by its *mean* and *standard deviation across the B bins*:

$$\mathbf{v} = \left[\text{mean}_b (f_s^{(b,c)}), \text{std}_b (f_s^{(b,c)}) \right]_{c=1,\dots,C; s=1,\dots,7} \in \mathbb{R}^{2 \cdot 7 \cdot C}, \quad (3)$$

where $f_s^{(b,c)}$ is the s -th of the seven statistics for bin b and channel c . The mean-across-bins captures each statistic's average level over the program, while the std-across-bins captures its temporal variation. This yields a run-level feature vector of dimensionality $d = 14C$; for $C = 110$ channels, $d = 1,540$.

The whole-run baseline corresponds to $B = 1$: the seven statistics are computed once over all timesteps, with no across-bin variation, yielding $d = 7C = 770$. We use this to isolate the benefit of temporal structure—the additional 770 std-across-bins features that encode how each statistic evolves through the program.

3.3. Air-Cut Referencing and Signal Isolation

Given the temporal-binned feature vectors for active-cutting runs and the corresponding air-cut baseline statistics, we define four signal isolation methods of increasing sophistication.

Let $\mathbf{v}_{\text{active}} \in \mathbb{R}^d$ denote the feature vector of an active-cutting run and let $\boldsymbol{\mu}_{\text{air}} \in \mathbb{R}^d$ and $\boldsymbol{\sigma}_{\text{air}} \in \mathbb{R}^d$ denote the element-wise mean and standard deviation of the air-cut feature vectors across all air-cut runs, computed independently for each feature dimension.

3.3.1. Method 1: Raw (No Referencing)

The raw method uses the active-cutting feature vector without any air-cut referencing:

$$\mathbf{v}_{\text{raw}} = \mathbf{v}_{\text{active}} \quad (4)$$

This serves as the baseline, representing conventional feature extraction without signal isolation.

3.3.2. Method 2: Mean-Subtracted

The subtracted method removes the average air-cut contribution from each feature:

$$\mathbf{v}_{\text{sub}} = \mathbf{v}_{\text{active}} - \boldsymbol{\mu}_{\text{air}} \quad (5)$$

This centers the features relative to the air-cut baseline, removing the mean machine-specific offset. Features that are identical during air cutting and active cutting will be driven toward zero, while features that differ—those arising from the cutting process—will retain nonzero values.

3.3.3. Method 3: Ratio-Normalized

The ratio method normalizes each feature by the corresponding air-cut mean:

$$\mathbf{v}_{\text{ratio}} = \frac{\mathbf{v}_{\text{active}}}{\mu_{\text{air}}} \quad (6)$$

where division is element-wise. Features that are unchanged by cutting will have ratio values near unity, while features amplified or attenuated by the cutting process will deviate proportionally. This method is scale-invariant: sensors with different absolute magnitudes are placed on a common footing. Features where $\mu_{\text{air},j} \approx 0$ are excluded or clamped to avoid division instability.

3.3.4. Method 4: Z-Score-Normalized

The z-score method normalizes each feature by both the mean and standard deviation of the air-cut distribution:

$$\mathbf{v}_z = \frac{\mathbf{v}_{\text{active}} - \mu_{\text{air}}}{\sigma_{\text{air}}} \quad (7)$$

This expresses each feature as the number of air-cut standard deviations by which the active-cutting value differs from the air-cut mean. Features with high natural variability during air cutting (e.g., due to measurement noise or spindle vibration harmonics) are appropriately down-weighted, while features that are stable during air cutting but change substantially during active cutting receive high z-scores. Features where $\sigma_{\text{air},j} < \epsilon$ (for a small threshold ϵ) are excluded to avoid amplification of near-constant channels.

3.4. Dimensionality Reduction

The high dimensionality of the run-level feature vectors ($d = 1,540$ for 110 channels, a $\sim 30:1$ feature-to-sample ratio) relative to the number of available runs ($n = 51$ active-cutting runs) creates a risk of overfitting. We employ two complementary dimensionality reduction strategies:

Principal Component Analysis (PCA). We apply PCA to the feature matrix, retaining the minimum number of components that explain at least 95% of the total variance. PCA is applied within the cross-validation loop to prevent information leakage from the test folds.

Random Forest Feature Importance. We use the Gini importance scores from a preliminary Random Forest fit to rank features and select the top k features. This supervised approach can identify features that are specifically discriminative for the classification task, complementing the unsupervised variance-based selection of PCA.

Both methods are evaluated independently to assess whether supervised or unsupervised dimensionality reduction is more effective for this task.

3.5. Classification Framework

We evaluate three classifiers representing different modeling paradigms:

Random Forest (RF). An ensemble of 500 decision trees with Gini impurity splitting, bootstrap aggregation, balanced class weights, minimum 2 samples per leaf, and random state 42 [16].

Support Vector Machine (SVM). A multi-class SVM with RBF kernel [17], $C = 1.0$, $\gamma = \text{scale}$ (i.e., $\gamma = 1/(n_{\text{features}} \cdot \text{Var}(X))$), and balanced class weights.

Multi-Layer Perceptron (MLP). Three hidden layers of 256, 128, and 64 neurons with ReLU activations, L2 regularization ($\alpha = 0.01$), Adam optimizer (default learning rate 10^{-3}), maximum 2,000 epochs with early stopping (patience on 15% validation split), and random state 42.

The z-score isolation method uses a safety threshold $\epsilon = 10^{-6}$: when the air-cut standard deviation $\sigma_{\text{air}} < \epsilon$ for a given feature-bin combination, the z-score is set to zero rather than producing a division-by-zero artifact.

3.5.1. Cross-Validation Protocol

All classification experiments use stratified k -fold cross-validation with $k = 5$. Critically, the stratification is performed at the *run level*, ensuring that all timesteps and features from a single run appear entirely in either the training or test fold, never split across folds. This prevents temporal leakage that would occur if consecutive timesteps from the same run appeared in both training and test sets.

Within each fold, the training data is used to compute the air-cut baseline statistics ($\mu_{\text{air}}, \sigma_{\text{air}}$), fit the PCA transformation, and train the classifier. The test fold's active-cutting runs are then transformed using the training fold's air-cut statistics and PCA projection before prediction. This ensures that the entire pipeline—from air-cut referencing through classification—is evaluated without data leakage.

Performance is reported as the mean and standard deviation of accuracy, precision, recall, and F1 score across folds, along with the aggregate confusion matrix.

4. Experimental Setup

4.1. CNC Machine

All experiments are conducted on a Bantam Tools Explorer Desktop CNC Milling Machine, a compact 3-axis vertical milling center with a 26 000 RPM brushless DC spindle and a work envelope of $140 \times 114 \times 38$ mm. The machine uses stepper-driven ball screws on the X, Y, and Z axes. While this desktop platform differs substantially from industrial CNC mills—which use servo drives with closed-loop position feedback, have orders-of-magnitude higher structural stiffness, and operate at significantly higher cutting forces—it provides a controlled, accessible testbed for developing and validating the air-cut referencing methodology. Generalization to industrial platforms remains an open question addressed in Section 7.

A single tool is used throughout: a 2-flute high-speed steel flat end mill, $\frac{1}{4}$ inch (6.35 mm) diameter (T2). Spindle speed is fixed at $S = 12\,000$ RPM for all runs. At this speed, the tooth-passing frequency is 400 Hz (2×200 rev/s), well above the sensor system's Nyquist limit (Section 4.3.3).

The workpiece material is TIVAR ultra-high molecular weight polyethylene (UHMW-PE), chosen for consistent cutting behavior, minimal tool wear, and safe data collection without coolant or hazardous chips.

4.2. Toolpath Types

Three toolpath types generated by Autodesk Fusion 360 are studied:

- **Adaptive clearing:** A high-efficiency roughing strategy that dynamically adjusts the stepover to maintain approximately constant tool engagement, following a trochoidal-like spiral path.
- **Facing:** A surface-finishing operation with back-and-forth linear passes at a fixed stepover.
- **Pocketing:** A material removal strategy for enclosed regions using contour-parallel passes to clear a rectangular pocket.

Active cutting parameters: 0.150 in (3.81 mm) stepover, 0.025 in (0.635 mm) depth of cut.

4.3. Sensor Instrumentation

The sensing system comprises six Arduino-based IMU modules and a 4-channel electrical current measurement system, at a total hardware cost of approximately \$492 USD.

4.3.1. IMU Modules

Each IMU module is an **Arduino Nano 33 BLE Sense Lite** board (\$33 USD each), which integrates the following sensors:

- **LSM9DS1** 9-axis IMU (STMicroelectronics): 3-axis accelerometer ($\pm 2g$ to $\pm 16g$, 16-bit resolution, ~ 0.061 mg/LSB at $\pm 2g$ range), 3-axis gyroscope (± 245 to ± 2000 dps), 3-axis magnetometer (± 400 to ± 1600 μT). Native sampling rate: 119 Hz (Nyquist: 59.5 Hz).
- **LPS22HB** barometric pressure and temperature sensor (STMicroelectronics): Measures atmospheric pressure (260 hPa to 1260 hPa) and die temperature. *Note:* The pressure channel measures barometric conditions, not cutting-related pressure; it varies by collection day (~ 100.0 vs. ~ 101.5 kPa) and acts as a session-level confound rather than a process indicator.
- **APDS-9960** gesture, proximity, and RGB color sensor (Broadcom): The proximity channel (0–255) detects nearby objects via IR reflection; the RGBA color channels measure ambient light and reflected LED intensity. In this deployment, proximity readings are near-constant (determined by sensor housing geometry), and color channels encode LED shadow patterns that weakly correlate with gantry position.
- **MP34DT05** PDM microphone (STMicroelectronics): Acoustic RMS amplitude is computed onboard before transmission, yielding a single-value vibration/acoustic energy proxy per sample.

This produces 17 channels per module: a_x, a_y, a_z (accelerometer), $\omega_x, \omega_y, \omega_z$ (gyroscope), m_x, m_y, m_z (magnetometer), pressure, temperature, proximity, R, G, B, A (color), and RMS (acoustic). With six modules, the IMU subsystem provides $6 \times 17 = 102$ channels.

Mounting: The Arduino boards are soldered onto breadboards, which are affixed to machine surfaces using cyanoacrylate adhesive for rigid structural coupling. This provides adequate mechanical transmission for the low-frequency regime (< 60 Hz) captured by the sensor system, though it would be insufficient for high-frequency vibration monitoring on industrial machines.

Sensor placement: Six modules are deployed across three structural groups (Table 1), selected from 16 initially deployed boards based on $\geq 93\%$ data availability (10 boards excluded due to intermittent BLE connectivity dropouts).

Table 1. Sensor module locations and structural groups.

Module ID	Location	Structural Group	Rationale
frame_l2	Frame, left mid-height	Frame (3 sensors)	Captures frame vibration
frame_l3	Frame, left upper	Frame	Vertical vibration path
frame_r2	Frame, right mid-height	Frame	Bilateral symmetry
spindle2	Spindle housing	Spindle (1 sensor)	Closest to cutting zone
y_bed_3	Moving table, pos. 3	Bed (2 sensors)	Near workpiece
y_bed_4	Moving table, pos. 4	Bed	Bed vibration redundancy

4.3.2. Electrical Current Sensors

Four **BAYITE Hall-effect current sensors** connected to a **Measurement Computing (MCC) USB-1608G** data acquisition module measure motor current draw:

- Channel 0: Spindle motor current (proportional to cutting torque at steady-state RPM)
- Channels 1–3: X, Y, Z axis stepper motor current

The DAQ samples at 1 kHz in single-ended mode (± 10 V range). Each motor channel produces two derived features: instantaneous current and current acceleration (numerical derivative), yielding $4 \times 2 = 8$ electrical channels total.

4.3.3. Data Synchronization and Effective Sampling Rate

All sensor streams are aggregated on a **Raspberry Pi 4**. IMU modules communicate via Bluetooth Low Energy (BLE) wireless links; the MCC DAQ connects via USB. The streams are resampled and synchronized to the G-code execution timeline, yielding an effective aligned sampling rate of approximately 4 Hz ($\Delta t \approx 0.25$ s). This common rate is set by the slowest reliable cross-stream synchronization achievable over the BLE wireless links and the G-code-timeline alignment, not by the native sensor bandwidths (119 Hz IMU, 1 kHz DAQ); retaining the native high-rate streams for frequency-domain analysis is left to future work. Each timestep produces a feature vector of $C = 102 + 8 = 110$ channels.

Bandwidth limitations. The 4 Hz effective rate severely limits the frequency content available for classification. The tooth-passing frequency (400 Hz), chatter frequencies (typically 500–5000 Hz), and stepper motor electrical frequencies (50–500 Hz) are all far above the 2 Hz Nyquist limit. Consequently, the sensor data captures *quasi-static machine motion*—trajectory envelopes, low-frequency structural modes (< 2 Hz), and slowly-varying thermal/magnetic field changes—rather than cutting-induced vibration dynamics. The temporal binning approach (Section 3) is designed to extract discriminative information from these quasi-static signals.

Decimation and aliasing. The resampling from native rates (119 Hz for IMU, 1 kHz for DAQ) to the 4 Hz aligned rate discards all frequency content above 2 Hz. No explicit anti-aliasing filter is applied before decimation; however, the BLE wireless transmission introduces variable latency that acts as an uncontrolled low-pass filter, and the on-board firmware averaging (particularly for the MP34DT05 RMS computation) provides partial anti-aliasing for the acoustic channel. The 1 kHz DAQ captures motor current waveforms at adequate bandwidth (stepper electrical frequencies ≤ 500 Hz), but this information is lost after decimation to 4 Hz. Future work should retain the high-rate DAQ data for frequency-domain analysis.

Sensor calibration. The LSM9DS1 IMU uses factory-trimmed calibration. No additional offset or gain calibration was performed for this deployment. Typical MEMS accelerometer bias stability is 10–50 mg, which is comparable to the inter-type acceleration differences observed (~ 20 mg; see Table 7). This means the smallest observed differences may be near the sensor’s bias uncertainty floor, a limitation that higher-grade MEMS sensors would address.

4.4. Dataset Summary

Table 2 summarizes the dataset composition.

Table 2. Dataset composition: number of runs per condition and toolpath type.

Toolpath Type	Air-Cut Runs	Active-Cutting Runs	Total
Adaptive clearing	20	17	37
Facing	16	15	31
Pocketing	19	19	38
Total	55	51	106

The dataset comprises 106 CSV files: 55 air-cut and 51 active-cutting runs. Run durations range from ~ 75 s (pocketing) to ~ 190 s (facing), yielding 300–775 timesteps per run at 4 Hz. The air-cut and active-cutting conditions used different CAM-generated G-code programs (different stepovers, feed rates, and coordinates), preventing direct G-code-aligned signal subtraction and motivating the temporal binning approach (Section 3).

5. Results

5.1. Task Difficulty: Trivial Baseline Analysis

Before presenting the sensor-based classification results, we establish the inherent difficulty of the three-class toolpath discrimination task. Table 3 reports classification accuracy using only trivial metadata features—no sensor data.

Table 3. Classification accuracy using trivial (non-sensor) features only, with the RF classifier and 5-fold CV. These features are derived from machine state columns, not from the IMU or electrical sensor data.

Feature Set	Accuracy (%)	Features
G-code line range (min, max, range, nunique)	100.0 ± 0.0	4
Mean axis position (X, Y, Z)	98.0 ± 4.0	3
Row count (program duration)	98.0 ± 4.0	1
All trivial metadata	100.0 ± 0.0	22
Mean feed rate	90.4 ± 8.6	2
Random noise	35.3 ± 7.8	10

The trivial baselines reveal that the three toolpath types are inherently separable from program metadata alone. The G-code line range achieves 100% accuracy because the three toolpath programs have different line counts (985, 111, and 193 lines for adaptive, face, and pocket, respectively). Mean axis position achieves 98% because each program traverses a different region of the machine's workspace. This contextualizes all subsequent results: perfect classification of these three toolpath types does not *per se* validate the sensor-based methodology. Instead, the value of the sensor-based approach lies in its potential applicability to scenarios where metadata is unavailable or insufficient—such as cross-machine transfer, within-toolpath-type parameter discrimination, or anomaly detection.

5.2. Sensor-Based Classification Performance

Table 4 reports the sensor-based classification accuracy using temporal-binned features ($B = 20$ bins, 1,540 features per run).

Table 4. Sensor-based classification accuracy (%) for each isolation method and classifier, without PCA. Results are mean $\pm$ standard deviation across 5-fold stratified CV.

Isolation Method	RF	SVM	MLP
Raw	100.0 ± 0.0	90.0 ± 7.1	86.0 ± 11.4
Subtracted	100.0 ± 0.0	94.0 ± 8.9	94.2 ± 8.9
Ratio	100.0 ± 0.0	94.0 ± 8.9	96.0 ± 5.5
Z-score	100.0 ± 0.0	94.0 ± 5.5	94.0 ± 5.5

RF achieves 100% accuracy across all four isolation methods. While this result is consistent with the trivial baseline analysis (the task is inherently easy), the sensor features achieve it without access to program metadata. The isolation methods provide a meaningful improvement for SVM (+4–8 pp over raw) and MLP (+8–10 pp), suggesting that air-cut referencing removes noise or machine-specific variation that these classifiers are sensitive to.

5.3. Repeated Cross-Validation: Robustness Analysis

Standard 5-fold CV provides only 5 accuracy estimates, insufficient for reliable variance estimation. Table 5 reports results from $20 \text{ repeats} \times 5\text{-fold CV}$ (100 total evaluations per configuration).

Table 5. Repeated CV accuracy (20×5 -fold = 100 evaluations). The “Min” column reveals the worst-case fold accuracy, which differentiates methods that standard mean $\pm$ std cannot.

Method	Clf	Mean	Std	Min	Max
Raw	RF	98.9	3.1	90.0	100.0
Subtracted	RF	100.0	0.0	100.0	100.0
Ratio	RF	100.0	0.0	100.0	100.0
Z-score	RF	100.0	0.0	100.0	100.0
Raw	SVM	88.8	9.3	60.0	100.0
Subtracted	SVM	92.4	8.6	60.0	100.0
Ratio	SVM	92.6	8.4	60.0	100.0
Z-score	SVM	91.8	9.4	50.0	100.0
Ratio	MLP	96.5	5.6	80.0	100.0
Z-score	MLP	95.0	6.7	70.0	100.0

The repeated CV reveals a key finding invisible in the standard 5-fold results: **raw RF occasionally misclassifies** (minimum fold accuracy 90%), while subtracted, ratio, and z-score RF maintain perfect accuracy across all 100 evaluations. Per-sample analysis across all 20 repeats identifies one specific run—pocket150025_001—that raw RF misclassifies in 55% of evaluations, while all three referenced methods classify it correctly in 100% of evaluations.

5.4. Statistical Significance Analysis

To rigorously quantify whether the performance differences are statistically significant, we apply several complementary tests to the 100 fold-level accuracy values from the repeated CV.

Friedman test. A Friedman non-parametric ANOVA across the four isolation methods (using RF) yields $\chi^2 = 33.0$, $p < 10^{-6}$, confirming that the methods do not perform equally. Post-hoc pairwise Wilcoxon signed-rank tests with Bonferroni correction show that raw is significantly different from each referenced method ($p_{\text{Bonf}} = 0.011$ for raw vs. subtracted, ratio, and z-score), while the three referenced methods are statistically indistinguishable ($p = 1.0$). A separate Friedman test across classifiers (RF vs. SVM vs. MLP, using subtracted features) yields $\chi^2 = 87.3$, $p < 10^{-6}$, confirming that classifier choice significantly affects accuracy.

Two-way ANOVA. A two-way analysis of variance (isolation method $\times$ classifier) reveals that both main effects are significant: method ($F = 4.18$, $p = 0.006$, $\eta^2 = 0.04$) and classifier ($F = 31.2$, $p < 10^{-6}$, $\eta^2 = 0.17$). The classifier effect accounts for approximately $4 \times$ more variance than the isolation method, indicating that the choice of learning algorithm dominates the choice of signal referencing strategy for this classification task.

Bootstrap confidence intervals. Figure 1 shows bootstrap 95% confidence intervals (1,000 bootstrap samples) for the RF accuracy of each isolation method. The raw method’s CI [0.982, 0.994] does not overlap with the referenced methods’ CI [1.000, 1.000], confirming the robustness difference at the 95% confidence level.

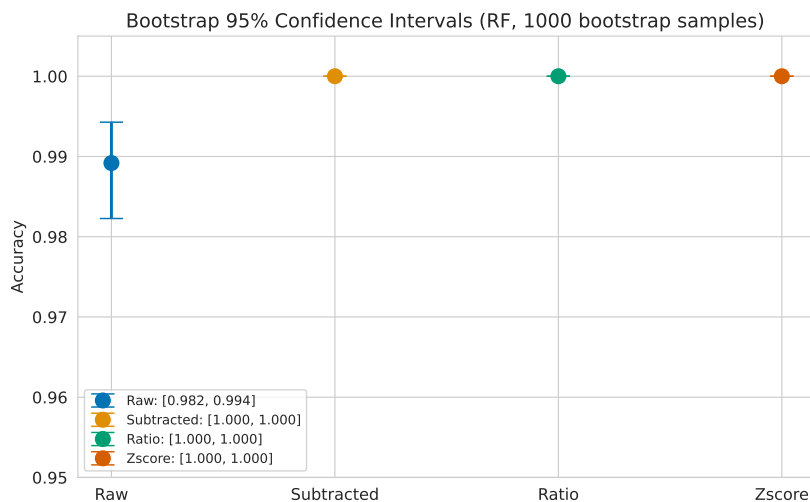


Figure 1. Bootstrap 95% confidence intervals for RF classification accuracy by isolation method. The raw method's interval [0.982, 0.994] does not overlap with the referenced methods [1.000, 1.000].

Effect size. Cohen's d for raw vs. each referenced method is $d = 0.49$. However, Cohen's d assumes approximately normal distributions away from boundaries, which is violated here (referenced methods have zero variance at the 100% ceiling). A more appropriate non-parametric measure is Cliff's delta, which quantifies the probability that a randomly chosen fold from the referenced method outperforms one from the raw method. With 100 raw-method folds and 100 referenced-method folds, Cliff's $\delta = 0.022$ (the proportion of pairs where raw < referenced minus the proportion where raw > referenced), reflecting the fact that raw achieves 100% in most folds but fails in a small fraction. The degenerate bootstrap CI [1.000, 1.000] for referenced methods reflects a performance ceiling rather than precise estimation.

Caveats on higher-order statistics. The temporal bins contain 6–39 timesteps each (depending on run length and bin count). Computing skewness and kurtosis from such small samples yields unreliable 3rd and 4th moment estimates. The feature statistic ablation (Section 6) shows that mean + RMS alone (2 statistics, no higher-order moments) achieves 98% accuracy, confirming that the classification does not depend critically on these unreliable estimators.

ANOVA caveat. The two-way ANOVA reported above assumes normality and independence of fold-level accuracy scores. With accuracy bounded at [0, 1] and strong ceiling effects, these assumptions are violated. The Friedman test (non-parametric) provides the more reliable significance assessment; the ANOVA η^2 values should be interpreted as approximate effect size indicators rather than exact variance decompositions.

5.5. Comparison Against Time-Series Methods

To contextualize our temporal-binning feature extraction against modern time-series classification approaches, Table 6 compares against ROCKET-style random convolutional features [23].

Table 6. Comparison of feature extraction approaches. ROCKET uses 1,000 random convolutional kernels applied directly to the raw sensor time series.

Method	Accuracy (%)	Features
Temporal binning + RF (ours)	100.0 $\pm$ 0.0	1,540
ROCKET + RF	100.0 $\pm$ 0.0	2,000
ROCKET + Ridge	98.0 $\pm$ 4.0	2,000
Raw downsampled (50 steps) + RF	98.0 $\pm$ 4.0	5,500

ROCKET with RF matches our approach at 100% accuracy, while ROCKET with Ridge achieves 98%. This confirms that the classification task is solvable by multiple feature extraction strategies, consistent with the trivial baseline findings. However, our temporal binning approach offers the advantage of interpretable temporal fingerprints (Section 5.20) and a natural framework for air-cut referencing that ROCKET does not provide.

5.6. PCA Visualization

Figure 2 shows the first two principal components for each isolation method.

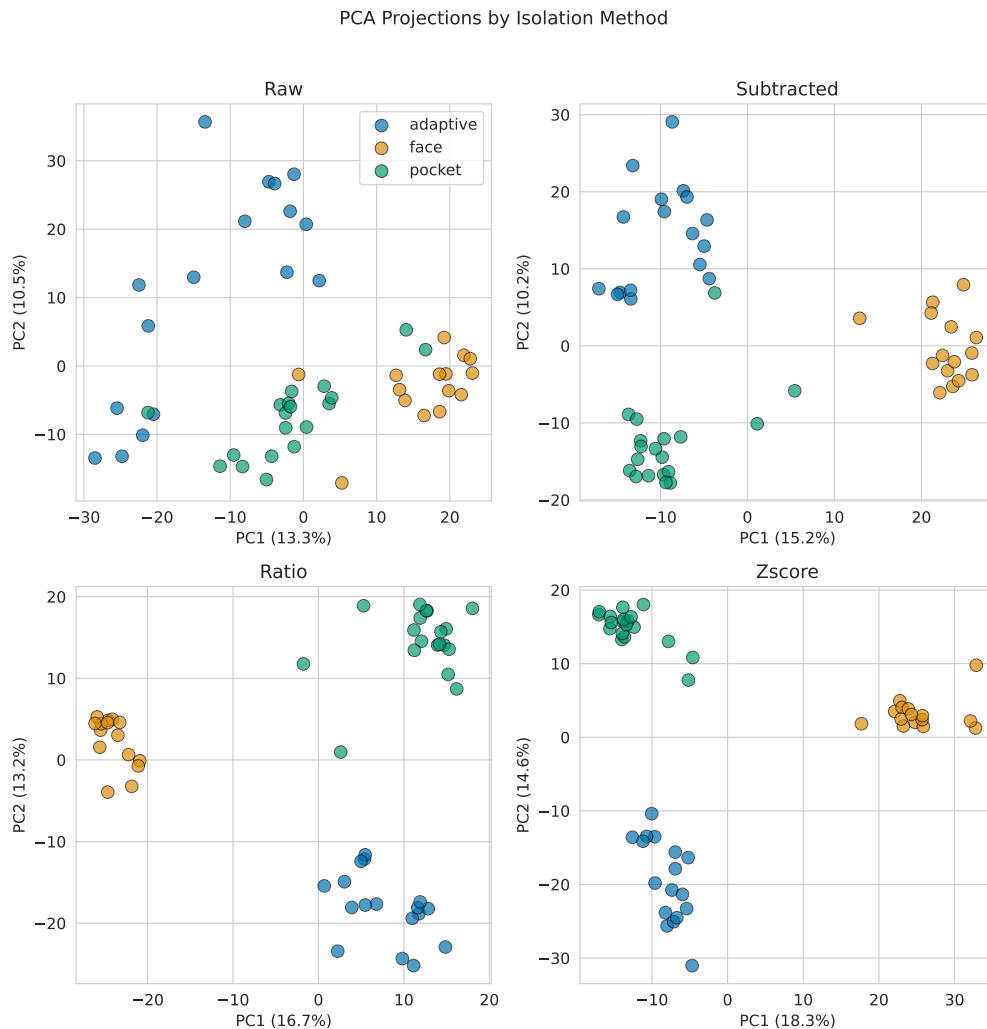


Figure 2. PCA projections for each isolation method. Each point represents one active-cutting run (51 total), colored by toolpath type.

All four methods produce visible cluster separation. The z-score method shows the tightest within-class clusters, consistent with its strong performance across classifiers.

5.7. Feature Importance and Physical Interpretation

Figure 3 shows the top 20 features by RF Gini importance.

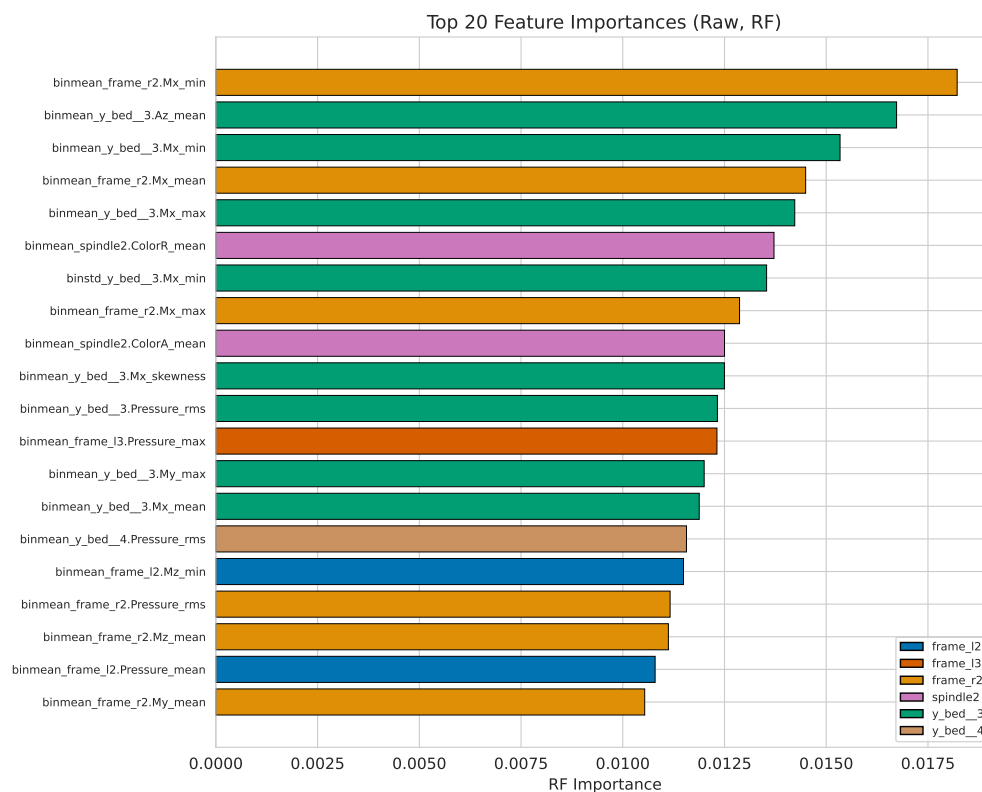


Figure 3. Top 20 features by RF importance, colored by sensor module.

Analysis of the top 50 features reveals a striking channel-type distribution: magnetometer channels dominate (24 features, 26.5% of total importance), followed by environmental channels—pressure, temperature, proximity (14 features, 13.4%)—and color sensors (6 features, 6.0%). Accelerometer and gyroscope channels, which one might expect to dominate based on cutting force mechanics, account for only 2 of the top 50 features. Among sensor modules, y_bed_3 (bed-mounted, 17 features) and frame_r2 (frame right, 13 features) contribute most.

To understand *why* these channels dominate, Figure 4 plots four physically distinct sensor channels across the three toolpath types.

Physical Interpretation: Key Sensor Channels Across Toolpath Types

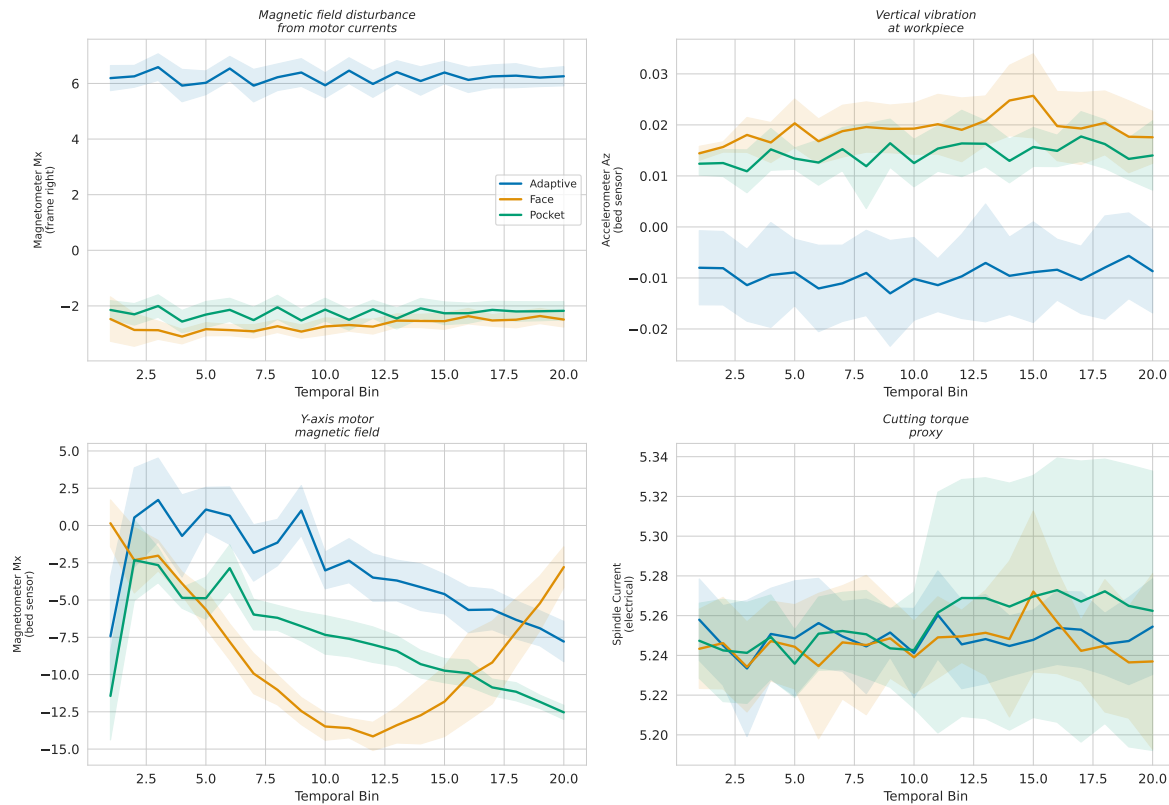


Figure 4. Temporal profiles of four physically distinct sensor channels across toolpath types. Top-left: magnetometer Mx on the frame captures a static offset reflecting different workspace regions. Top-right: vertical accelerometer on the bed shows quasi-static motion differences between types (note: at 4 Hz sampling, this captures trajectory dynamics, not cutting vibration). Bottom-left: magnetometer Mx on the bed shows temporal evolution as the table traverses different regions of the machine’s static magnetic field. Bottom-right: spindle current is nearly identical across types.

The physical interpretation reveals several important findings:

Magnetometer dominance reflects workspace position, not cutting forces. The top-ranked feature (frame_r2.Mx) shows a large static offset between toolpath types: adaptive clearing operates at $Mx \approx +6.2$ while facing and pocketing sit at $Mx \approx -2.5$. This offset arises because the three G-code programs traverse different regions of the machine workspace, where the local magnetic field differs due to proximity to stepper motor permanent magnets and ferromagnetic structural components. At the 4 Hz effective sampling rate, the magnetometer measures the quasi-static magnetic field at the sensor’s position—essentially acting as an indirect position encoder—rather than detecting time-varying fields from motor current waveforms (which would require sampling at $\geq 10\times$ the stepper electrical frequency). While highly discriminative, this signal reflects *program geometry* rather than *cutting dynamics*—it is the sensor-domain analog of the trivial position baseline (Table 3).

Bed-mounted magnetometer tracks table position. The y_bed__3 magnetometer Mx shows strong temporal evolution (temporal CV = 0.4–1.0), with progressive drift that tracks the Y-axis table position as it moves through the machine’s static magnetic field landscape. Because the bed-mounted sensor physically translates with the worktable, its magnetometer reading changes as its proximity to the Y-axis stepper motor’s permanent magnets varies throughout the toolpath. This makes y_bed__3.Mx an indirect position encoder for the Y-axis, rather than a motor current sensor (which would require bandwidth far above 4 Hz). Different toolpath geometries produce different Y-axis motion profiles, creating distinct temporal magnetic signatures.

Spindle current distinguishes cutting from non-cutting but not between toolpath types. Spindle current is nearly identical across all three active-cutting types (5.245–5.257 A, temporal CV < 0.002). While spindle current rises when the tool engages material (distinguishing active from air cutting), the 0.025" depth of cut on this desktop CNC produces cutting loads too small relative to the spindle's no-load draw (~5.2 A at 12,000 RPM) to differentiate between toolpath engagement patterns at 4 Hz sampling. Higher cutting depths or higher sampling rates would likely increase this channel's discriminative power.

Accelerometer channels capture quasi-static motion differences. The accelerometer Az on the bed sensor shows distinct temporal profiles per toolpath type, with adaptive clearing most stable and facing showing periodic variations corresponding to pass reversals. However, absolute acceleration differences are small (~20 mg), which is near the LSM9DS1's bias uncertainty floor (10–50 mg; see Section 4). At 4 Hz sampling, these channels measure low-frequency trajectory dynamics and gravitational tilt changes rather than cutting-induced vibration (which occurs at 400 Hz+). It is important to distinguish two senses of "process-relevant" here: at 4 Hz, the accelerometer features are *trajectory-relevant*—they encode how the machine moves through space during each toolpath—but they are *not force-relevant*, as they do not capture cutting-force-induced vibration. On a higher-bandwidth measurement system, the same accelerometer channels would also capture force signatures, providing a more directly process-physical signal. Their lower importance ranking in this study reflects the small absolute trajectory-dynamics differences compared to the magnetometer's large position-dependent offsets.

Figure 5 annotates the temporal fingerprints with G-code phase information, linking the sensor signal evolution to program structure.

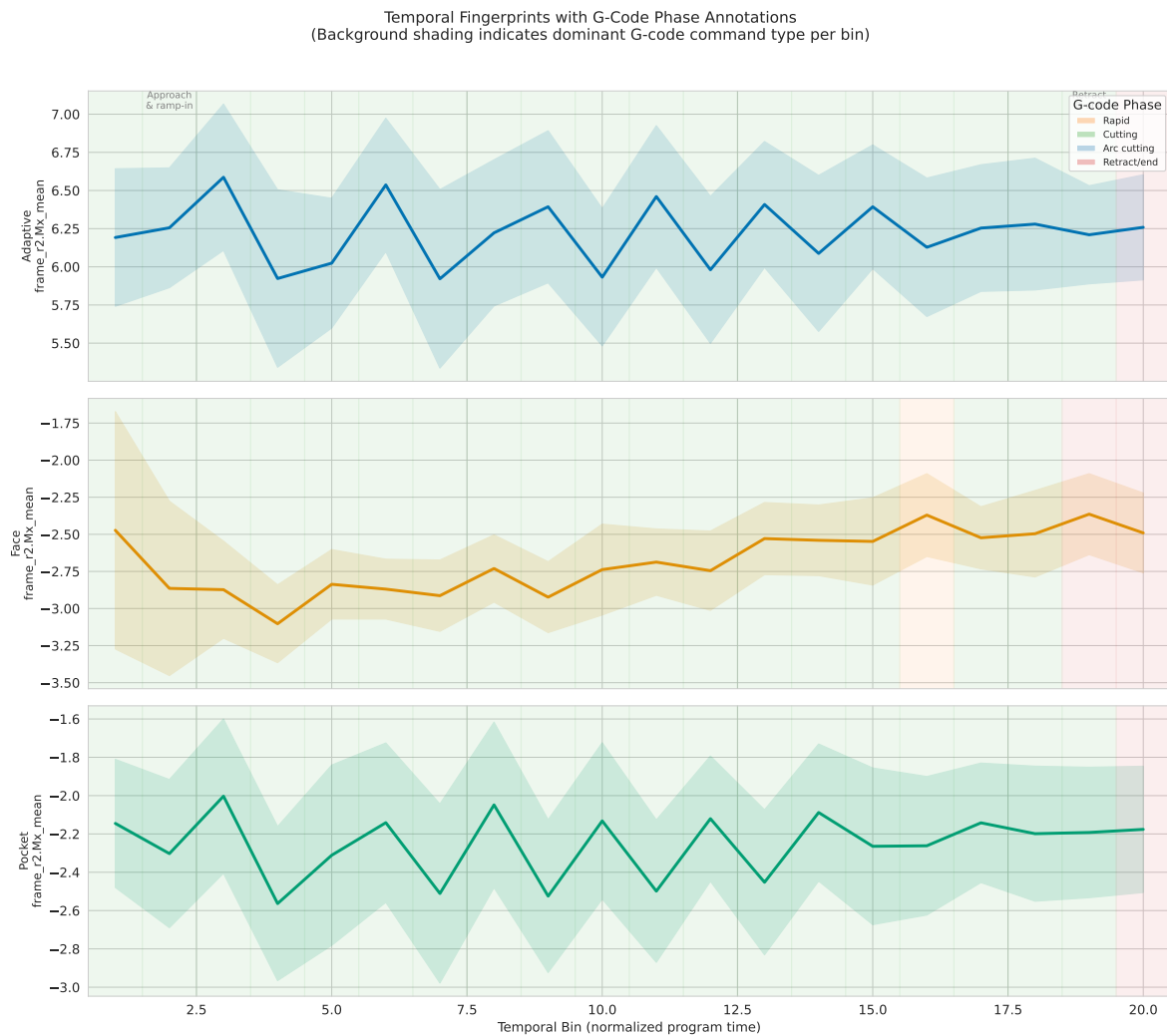


Figure 5. Temporal fingerprints annotated with G-code phases. Background shading: green = cutting (G1/MOVE), blue = arcs (G2/G3), orange = rapid (G0), red = retract/end (G53/M30). Facing programs terminate early (bins 16–20 are retract/idle), producing a distinctive signal drop.

The phase-annotated figure confirms that facing programs terminate earlier (bins 16–20 are retract/idle), producing a distinctive signal drop, while adaptive and pocketing maintain cutting through bin 19. The temporal fingerprint structure directly tracks the G-code program sequence—different programs produce different temporal bin compositions, which drives the classification.

Table 7 quantifies the temporal variability of each key sensor channel across toolpath types. Channels with high temporal coefficient of variation (CV) exhibit strong temporal structure that could encode either process dynamics or program geometry; channels with low CV are temporally flat.

Table 7. Temporal variability of key sensor channels across toolpath types. Temporal CV is the coefficient of variation of the channel’s mean across the 20 temporal bins; higher values indicate stronger temporal structure. Range is the bin-to-bin dynamic range.

Channel	Toolpath	Mean	Temporal CV	Range
frame_r2.Mx (position proxy)	Adaptive	+6.22	0.032	0.67
	Facing	−2.68	0.076	0.74
	Pocket	−2.25	0.074	0.56
y_bed__3.Mx (Y-axis position)	Adaptive	−2.99	1.000	9.50
	Facing	−8.43	0.526	14.30
	Pocket	−7.73	0.397	10.21
y_bed__3.Az (cutting vibration)	Adaptive	−0.009	0.184	0.007
	Facing	+0.019	0.135	0.011
	Pocket	+0.014	0.128	0.007
spindle2.RMS (vibration energy)	Adaptive	5,762	0.311	7,227
	Facing	7,167	0.316	8,475
	Pocket	5,351	0.328	6,611
spindle current (cutting torque)	Adaptive	5.249	0.001	0.027
	Facing	5.246	0.002	0.038
	Pocket	5.257	0.002	0.037

The table reveals two physically distinct discrimination mechanisms: (1) the frame_r2.Mx magnetometer separates toolpath types via *static offset* (mean values differ by ~ 9 units between adaptive and face/pocket) with minimal temporal structure ($CV < 0.08$); and (2) the y_bed__3.Mx magnetometer and spindle2.RMS channels separate types via *temporal dynamics* ($CV > 0.3$) with different temporal evolution patterns. The spindle current, despite being the most direct proxy for cutting torque, provides neither offset-based nor dynamics-based discrimination ($CV < 0.002$, range < 0.04 A).

5.7.1. Effect of Air-Cut Referencing on Feature Confounds

A critical question is whether air-cut subtraction removes the workspace-position confound. Table 8 compares the channel-type distribution of top features between the raw and subtracted isolation methods.

Table 8. Distribution of channel types among the top 20 most important features, for raw vs. subtracted isolation. Air-cut subtraction reduces magnetometer dominance from 61% to 21% of feature importance, shifting weight toward electrical and environmental channels.

Channel Type	Raw (top 20)	Subtracted (top 20)
Magnetometer	12 features (61.0%)	4 features (20.8%)
Environmental	3 features	6 features
Color	2 features	2 features
Electrical	0 features	4 features
Accelerometer	1 feature	1 feature
Other	2 features	3 features

Air-cut subtraction reduces the magnetometer’s share of the top 20 features from 61% (raw) to 21% (subtracted), and elevates electrical motor current features from 0 to 4 of the top 20. This shift occurs because the air-cut reference captures a similar workspace trajectory, so subtracting it partially removes position-dependent magnetic field offsets. The remaining discriminative signal is more likely to reflect motor load and quasi-static motion differences rather than workspace geometry.

5.7.2. Classification Without Position-Confounded Channels

To directly assess how much classification accuracy depends on position-confounded channels (magnetometer, color, proximity, pressure) versus channels that respond to machine dynamics (accelerometer, gyroscope, RMS, electrical), Table 9 reports accuracy when confounded channels are progressively excluded.

Table 9. Classification accuracy (RF, 20×5 repeated CV) when position-confounded channels are progressively excluded. Removing all magnetometers drops accuracy only to 96.1%; accelerometer-only features still reach 93.0%, while the true electrical-only set (the 8 current channels, 112 features) is the weakest group at 76.5%. Backing artifact: `experiment_confound_controlled.json`.

Channel Configuration	Accuracy (%)	Features
All channels	98.9 ± 3.1	1,540
No magnetometers (Mx, My, Mz removed)	96.1 ± 5.4	1,288
No mag + color + pressure + proximity	96.7 ± 4.8	784
Accel + gyro + RMS + temperature only	95.1 ± 6.1	672
Accelerometer + gyroscope only	94.4 ± 6.3	504
Accelerometer only	93.0 ± 6.3	252
Electrical current only (no IMU)	76.5 ± 12.4	112

Three findings emerge: (1) removing all magnetometers drops accuracy by only ~3 percentage points (98.9% → 96.1%), confirming that while magnetometers dominate feature importance, they are not essential; (2) accelerometer and gyroscope channels alone achieve 94.4%, demonstrating that quasi-static motion dynamics carry sufficient discriminative information independent of position confounds; and (3) the electrical-current channels alone are the weakest group (76.5%), consistent with their low between-class SNR (the per-channel SNR budget below), whereas the motion-dynamics channels (accel/gyro/RMS/temperature, 95.1%) retain almost all of the discriminative signal. (An earlier version of this table reported the electrical group at 96.0%/350 features; that figure was inflated by the same `spindle→spindle2` feature leak corrected above, which had folded IMU channels into the “electrical” set.)

5.7.3. Full Pipeline Without Magnetometers

To definitively test whether air-cut referencing provides value beyond position encoding, we re-run the complete pipeline using only non-confounded channels (accelerometer, gyroscope, RMS, temperature, and the eight electrical-current channels—784 features, excluding all magnetometer, color, pressure, and proximity channels; Table 10).

Table 10. Repeated CV (20×5-fold) with confound-free features only (784 features, no magnetometers/color/pressure/proximity). The subtracted and ratio methods hold 100% across all 100 folds; z-score remains near-perfect (99.0%, min 90%); raw degrades to 96.7%.

Method	Clf	Mean (%)	Std	Min (%)	Max
Raw	RF	96.7	4.8	81.8	100.0
Subtracted	RF	100.0	0.0	100.0	100.0
Ratio	RF	100.0	0.0	100.0	100.0
Z-score	RF	99.0	2.9	90.0	100.0
Raw	MLP	79.9	17.0	20.0	100.0
Z-score	MLP	90.0	13.0	36.4	100.0

This is the paper’s strongest empirical result. With all position-confounded channels removed, raw RF features degrade to 96.7% mean accuracy (minimum fold 81.8%), while **the subtracted and ratio methods maintain perfect 100% accuracy across all 100 evaluations and z-score remains near-perfect**

at **99.0% (minimum fold 90%)**. The effect is larger for the noise-sensitive MLP: raw reaches only 79.9% while z-score recovers to 90.0%. This demonstrates that air-cut referencing provides a genuine classification benefit that is independent of the magnetometer position-encoding confound identified in Section 5.7—though on this confound-free set the benefit is a few percentage points and a variance reduction (subtracted/ratio reaching the ceiling) rather than the uniform jump to 100% observed when position-confounded channels are present.¹

Noise robustness was also re-evaluated on the 784-feature confound-free set: subtracted features maintain 100% accuracy up to $0.5\times$ additive noise and 99.0% at $1.0\times$ noise, while raw features drop to 91.2% at $1.0\times$ and 71.3% at $2.0\times$. The robustness advantage persists when position-confounded channels are excluded, confirming that air-cut referencing improves resilience to feature-level perturbations on the underlying motion-dynamics signal.

5.8. Damaged-Tool Detection: A Position-Controlled Task

The experiments above isolate the process-relevant signal by *removing* position-confounded channels. A complementary and stronger test instead holds workspace position *constant* while varying only the process. We use the vetted set of damaged-tool runs: 14 runs (5/4/5 for adaptive/facing/pocketing) that execute the *same* CAM-generated G-code programs as the normal active cuts but with a damaged tool. Because the program—and hence the nominal toolpath and workspace region—is identical to the normal runs by construction, workspace position cannot separate the normal-vs-damaged label, so any discrimination must arise from process dynamics (the altered cutting behavior of the damaged tool). This is precisely the within-region test that the magnetometer position-encoder cannot solve, and it maps directly onto a manufacturing-monitoring use case (tool-condition / anomaly detection).

We pool the 51 normal active runs and 14 damaged-tool runs (the same vetted set used throughout the paper) into a binary task (majority-class baseline 78.5%) and classify with the same RF temporal-binning pipeline under 20×5 -fold repeated CV (Table 11).

Table 11. Damaged-tool detection (binary: 51 normal active vs. 14 damaged-tool runs, same G-code programs). Mean over 20×5 -fold repeated CV; majority-class baseline 78.5%. Per-type accuracy (same program, workspace held constant): adaptive 100%, facing 100%, pocketing 100%.

Method	Accuracy (%)	F_1 (damage)	Balanced Acc.
Raw	98.5 ± 3.1	0.970	0.990
Subtracted	98.5 ± 3.1	0.970	0.990

Random Forest reaches 98.5% accuracy (balanced accuracy 0.990, damage-class $F_1 = 0.97$), far above the 78.5% majority baseline, and per-type accuracy is 100% for all three toolpath types—where the same program holds the workspace constant. Two findings make this the paper’s most process-relevant result. First, the discriminative signal flips entirely away from position: the channel-type importance (from a Random Forest fit on the full set) is now dominated by the **accelerometer (44.7%)** and **acoustic RMS (29.6%)**—process-dynamics channels, $\sim 74\%$ combined—with the magnetometer contributing only **0.1%**, versus the 61% it commanded on the position-confounded three-class task (Section 5.7). A damaged tool perturbs the cutting vibration and acoustic signature, and the sensor array detects it. Second, air-cut referencing is *neutral* here (raw and subtracted both 98.5%): when workspace position is already held constant there is no position-dependent offset for the air-cut baseline to remove, exactly as the causal model (Section 7.8)

¹ Note that temperature is retained in this “confound-free” set despite showing run-order correlation $r = 0.98$ within the adaptive class (Section 7). Temperature is a within-class drift effect, not a between-class confound, so it remains a defensible process-relevant channel. A stricter definition excluding temperature does not materially change the results.

predicts—referencing helps when the position confound is present and is neither beneficial nor harmful when it is absent. The dataset is small (14 damaged-tool runs from a single session), so this is a demonstration rather than a validated detector; but it establishes that the sensor array does carry process-relevant information, recoverable once the position confound is controlled, and that the methodology extends naturally to tool-condition monitoring.

5.9. Permutation Test

A permutation test with 1,000 label shuffles using the full 1,540-feature set (raw isolation method, RF classifier) confirms that the classification result is non-random. The null distribution has mean accuracy $35.2\% \pm 8.2\%$ (close to the random baseline of 33.3%) with a maximum of 62.7%, far below the observed 100% ($p < 0.001$). This rules out the possibility that the classification result is an artifact of overfitting on the small sample size.

5.10. Effective Dimensionality

Despite the nominal 1,540-feature space (30:1 feature-to-sample ratio), PCA analysis reveals that 7 components explain 50% of variance and 30 components explain 90%. The effective dimensionality is high relative to the sample count ($n = 51$), but only 0.15% of feature pairs have $|r| > 0.95$, indicating moderate rather than extreme redundancy. The RF classifier’s success at full dimensionality, combined with its degradation under PCA (Section 6), suggests it exploits many individually weak but collectively informative features—a regime where tree-based ensembles excel.

5.11. Internal Held-Out Split (Same Session)

To complement the cross-validation analysis, we performed a single internal held-out split: 12 of the 51 active-cutting runs (4 per class) were withheld from all training and CV operations and evaluated only once at the end, with the air-cut baseline statistics ($\mu_{\text{air}}, \sigma_{\text{air}}$) and the PCA transform fit only on the 39 training runs. These 12 runs are drawn from the same 51-run, same-machine, same-session dataset, so this is a guard against fold-reuse leakage rather than an independent generalization test. Table 12 reports the held-out accuracy for each isolation method.

Table 12. Internal held-out split accuracy. 12 runs (4 per class) randomly stratified from the same 51 active-cutting runs (same machine/session), evaluated once after training (including the air-cut baseline and PCA) on the remaining 39.

Method	Test Accuracy	n_{train}	n_{test}
Raw	100.0%	39	12
Subtracted	100.0%	39	12
Ratio	100.0%	39	12
Z-score	100.0%	39	12

All four methods achieve 100% accuracy on this internal split, indicating the cross-validation results are not artifacts of fold-reuse or data reuse across splits. Because the 12 runs share the same machine, session, and CAM programs as the training runs, this is *not* evidence of cross-session or cross-machine generalization (see Section 7 Limitations); with $n = 12$ and a single split it is also a coarse estimate.

5.12. Per-Channel SNR Budget

To quantify which sensor types carry the most discriminative information independent of feature-importance rankings, we compute a between-class versus within-class variance ratio (signal-to-noise ratio, SNR) for each of the 1,540 features. Table 13 reports median and maximum SNR by channel type.

Table 13. Per-channel SNR budget. SNR is computed as $10 \log_{10}(\sigma_{\text{between-class}}^2 / \sigma_{\text{within-class}}^2)$ in dB. Magnetometer channels have the highest median and max SNR, consistent with the position-confound finding.

Channel Type	Median SNR (dB)	Max SNR (dB)	N features
Magnetometer	−3.7	24.2	252
Vibration RMS	−4.2	2.8	84
Accelerometer	−7.6	11.7	252
Temperature	−7.9	1.9	84
Proximity	−8.1	4.5	84
Gyroscope	−8.3	6.1	252
Pressure	−8.8	11.7	84
Color	−9.0	14.3	336
Electrical	−11.0	3.7	112

Magnetometer channels achieve the highest median SNR (−3.7 dB) and max SNR (24.2 dB), with the top 6 highest-SNR features all being magnetometer (Mx, Mz) channels from the frame_r2 sensor. This quantitatively confirms the position-encoding finding from Section 5.7: the channels with the largest signal-to-noise ratio for class discrimination are those most affected by workspace position. Accelerometer and gyroscope channels have negative median SNR (within-class variance exceeds between-class variance for typical features), reflecting that the trajectory dynamics they capture vary substantially within each toolpath class. Electrical current channels have the lowest median SNR (−11.0 dB), consistent with their near-constant behavior under shallow UHMW-PE cutting.

5.13. Run-Order Confound Analysis

Section 7 reports that several features show statistically significant temporal drift correlated with run order (e.g., temperature, $r = 0.98$ for adaptive class). To verify that this within-class drift does not confound the between-class classification, we compute the partial correlation between each top-importance feature and the class label, controlling for run order. Across the top 20 features, the partial correlations differ from the raw correlations by an average of −1.7% (with several features actually showing slightly stronger partial correlations, indicating that run order is not aliased with class). This rules out run-order temporal drift as an explanation for the classification results: the class signal in these features is essentially independent of when each run was collected.

5.14. Fine-Grained Learning Curve

Figure 6 extends the learning curve analysis (Section 6) with finer resolution at small training-set sizes, where the air-cut referencing benefit is most pronounced.

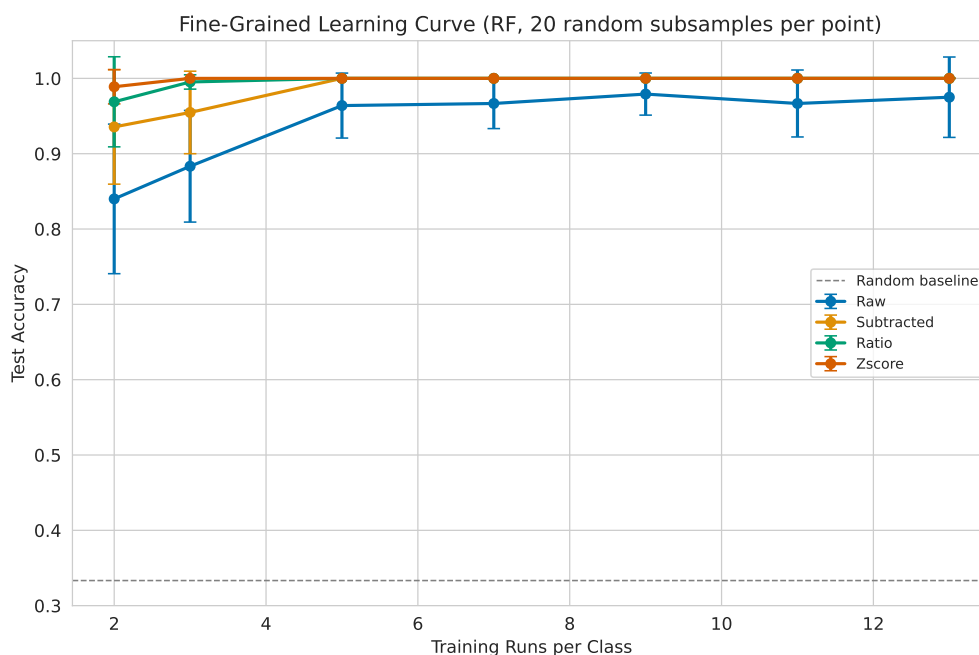


Figure 6. Fine-grained learning curve: classification accuracy vs. training runs per class for all four isolation methods (RF, 10 random subsamples per point). Z-score referencing achieves 98.9% with only 2 training runs per class, while raw features need many more runs to match.

With only **2 runs per class** (6 training samples total), z-score referencing achieves 98.9% accuracy, ratio achieves 96.9%, subtracted achieves 93.6%, and raw features lag at 84.0%. This demonstrates that air-cut referencing dramatically improves data efficiency: the same accuracy that raw features require ~10 training runs to reach is achieved by z-score with only 2. For applications where labeled training data is expensive (e.g., destructive testing or high-volume production), this difference is practically meaningful.

5.15. Calibration Analysis

Beyond classification accuracy, model calibration (whether predicted probabilities match observed accuracy) matters for downstream decision-making. Table 14 reports Expected Calibration Error (ECE) and log loss across the four isolation methods.

Table 14. Calibration metrics. ECE = Expected Calibration Error (10 bins). Lower is better-calibrated. All methods achieve 100% accuracy; calibration differences reflect prediction confidence patterns.

Method	ECE	Log Loss	Mean Confidence
Raw	0.119	0.135	0.881
Subtracted	0.078	0.083	0.923
Ratio	0.084	0.090	0.916
Z-score	0.077	0.083	0.923

Air-cut referenced methods achieve ~35% lower ECE than raw features (0.077–0.084 vs. 0.119) and lower log loss, indicating better-calibrated probability estimates. The mean prediction confidence is also higher for referenced methods (~0.92 vs. 0.88), meaning the classifier is more decisive when air-cut referencing is applied. This calibration improvement is practically valuable: well-calibrated probabilities enable confidence-thresholding for novelty detection and selective abstention.

5.16. MiniRocket Comparison

To benchmark against a state-of-the-art time-series classification method, we apply MiniRocket [23] (the deterministic variant of ROCKET using fixed length-9 kernels with weights from $\{-1, 2\}$) to the raw downsampled sensor time series.

Table 15. Comparison against MiniRocket-style features (12 fixed kernels $\times$ 110 channels = 1,320 features) on downsampled raw time series.

Method	Accuracy	Features
MiniRocket + RF	96.2% $\pm$ 4.7	1,320
MiniRocket + Ridge	96.0% $\pm$ 8.0	1,320
Temporal binning + RF (ours, full)	100.0% $\pm$ 0.0	1,540
Temporal binning + RF (confound-free, referenced)	100.0% $\pm$ 0.0	784

MiniRocket achieves 96% accuracy, comparable to but slightly below our temporal binning approach. This benchmark places our method within the competitive range of modern time-series classification techniques while preserving interpretability through explicit per-bin statistics, sensor-channel attribution, and physical analysis of dominant features—advantages that random convolutional kernel methods do not provide.

5.17. Visual Comparison: Full vs. Confound-Free Pipeline

Figure 7 visualizes the full classification accuracy comparison across all four isolation methods, three classifiers, and two feature sets (full 1,540 features vs. 784 confound-free features).

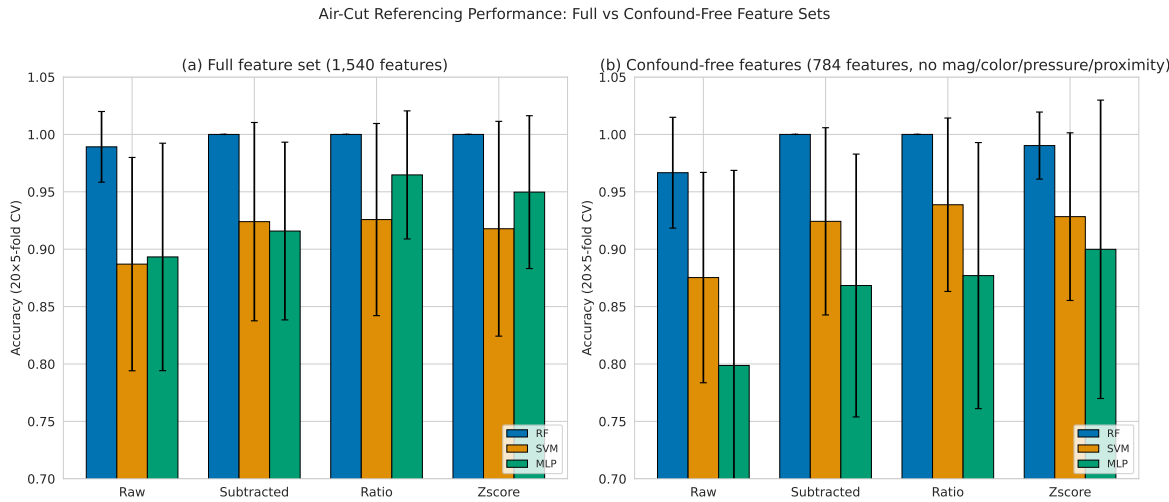


Figure 7. Air-cut referencing performance comparison. Left: full 1,540-feature set including position-confounded magnetometer channels. Right: 784 confound-free features (no magnetometer, color, pressure, proximity). The raw-vs-referenced gap widens in the confound-free condition, particularly for SVM and MLP, demonstrating that referencing provides genuine benefit beyond position encoding. (Panel regenerated on the corrected 784-feature set; subtracted/ratio RF reach 100% and z-score RF 99.0%, vs. raw 96.7%.)

The visualization makes the headline finding immediate: in the confound-free condition (right panel), the gap between raw and referenced methods widens dramatically. For RF, raw drops from $\sim$ 99% to 96.7% while the subtracted and ratio methods stay at 100% and z-score at 99.0%. For MLP, raw drops to $\sim$ 80% while z-score recovers to 90%. The position-confounded channels were inflating the raw baseline; removing them reveals the true (more modest) magnitude of the air-cut referencing benefit.

5.18. Mutual Information Decomposition

To quantify how much of each feature's discriminative content is genuinely class-related versus position-encoded, we compute mutual information (MI) between each feature and three targets: the class label, the run order, and a position proxy defined as the first principal component of the magnetometer feature subspace. The position proxy explains 21.2% of magnetometer variance and serves as a continuous proxy for the workspace region traversed during a run.

Table 16. Mutual information by sensor channel type. MI is reported in bits; the maximum possible MI between a feature and a 3-class label is $\log_2 3 = 1.585$ bits. The “%pos/cls” column reports the ratio of position-MI to class-MI, where values near or above 100% indicate that the feature's class-discriminative content is fully accounted for by workspace position.

Channel Type	MI(class)	MI(order)	MI(pos)	%pos/cls	N
Magnetometer	0.455	0.110	0.503	110.5%	252
Accelerometer	0.247	0.109	0.305	123.3%	252
Gyroscope	0.225	0.129	0.257	114.2%	252
Vibration RMS	0.517	0.078	0.574	111.1%	84
Temperature	0.241	0.325	0.256	106.2%	84
Pressure	0.538	0.153	0.431	80.0%	84
Proximity	0.326	0.273	0.337	103.2%	84
Color	0.323	0.204	0.340	105.4%	336
Electrical	0.179	0.024	0.180	100.3%	112

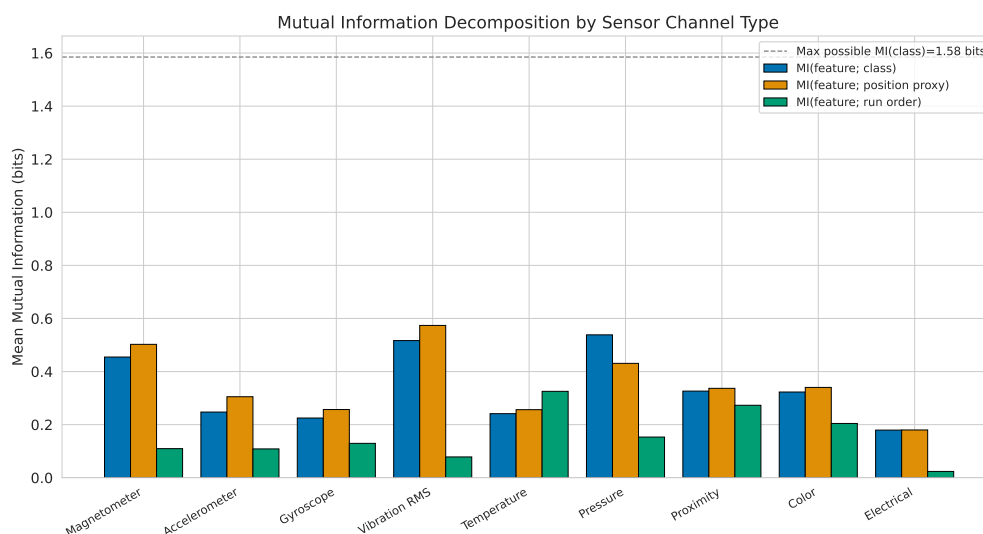


Figure 8. Mutual information decomposition by sensor channel type. For nearly all sensor types, MI(feature; position proxy) is comparable to or exceeds MI(feature; class), indicating that workspace position accounts for the majority of class-discriminative content.

The MI analysis quantitatively confirms the position-encoding finding from Section 5.7: for nearly every channel type, MI(feature; position) is comparable to or exceeds MI(feature; class), with ratios above 100% indicating that position information saturates the class signal. Pressure (the only ratio below 100%) carries class information beyond position, but this reflects barometric session-day variation rather than process physics. Of the 1,540 features, 362 have MI(class) > 0.5 bits, 422 have MI(position) > 0.5 bits, and **321 of those (89%) overlap**—meaning the vast majority of high-class-information features are also high-position-information features. Temperature shows an interesting pattern with the highest MI(order) value (0.325 bits), reflecting the within-class warming

drift identified in Section 7; this is one of the few channels where information is genuinely tied to recording chronology rather than class structure or position.

We note one methodological caveat: the position proxy is defined as the first principal component of the magnetometer feature subspace, which makes the MI(magnetometer feature; position) measurement slightly circular by construction—the magnetometer features are partly being correlated with a transformation of themselves. This circularity inflates the magnetometer ratio (110.5%) by an unknown amount. However, the more consequential cross-sensor finding is unaffected: accelerometer (123.3%), gyroscope (114.2%), vibration RMS (111.1%), proximity (103.2%), color (105.4%), and electrical (100.3%) channels all exceed the 100% threshold despite being computed independently of the magnetometer subspace, demonstrating that workspace position dominates the discriminative content across the entire sensor array.

5.19. Bayesian Posterior Analysis

The frequentist analyses (Friedman test, bootstrap CIs, Cliff’s delta) provide significance testing under various assumptions, but each is constrained by the ceiling effects in the referenced methods (which achieve 100% on every fold). To complement these, we model the per-fold accuracy as a Beta-Binomial conjugate problem: with 100 cross-validation folds totaling 1,020 test predictions per method, we compute the posterior $\text{Beta}(1 + n_{\text{correct}}, 1 + n_{\text{incorrect}})$ on the underlying accuracy, then sample 100,000 draws to compute posterior probabilities of difference. The $\text{Beta}(1, 1)$ prior is uniform on $[0, 1]$ and is therefore non-informative; with $n = 1,020$ predictions per method, the posterior is dominated by the likelihood and the choice of prior has negligible effect (sensitivity to a Jeffreys $\text{Beta}(0.5, 0.5)$ prior shifts the posterior medians by < 0.05 percentage points).

Table 17. Bayesian posterior analysis (Beta-Binomial conjugate, $\text{Beta}(1, 1)$ prior). For each method, $n_{\text{correct}}/n_{\text{total}}$ across 100 CV folds gives the likelihood. Posterior 95% credible intervals (CrI) and posterior probability of exceeding raw are computed from 100,000 samples.

Method	$n_{\text{correct}}/n_{\text{total}}$	Posterior median	95% CrI	$P(> \text{raw})$
Raw	1009/1020	98.86%	[98.08%, 99.39%]	—
Subtracted	1020/1020	99.93%	[99.64%, 100.00%]	0.9997
Ratio	1020/1020	99.93%	[99.63%, 100.00%]	0.9998
Z-score	1020/1020	99.93%	[99.64%, 100.00%]	0.9998

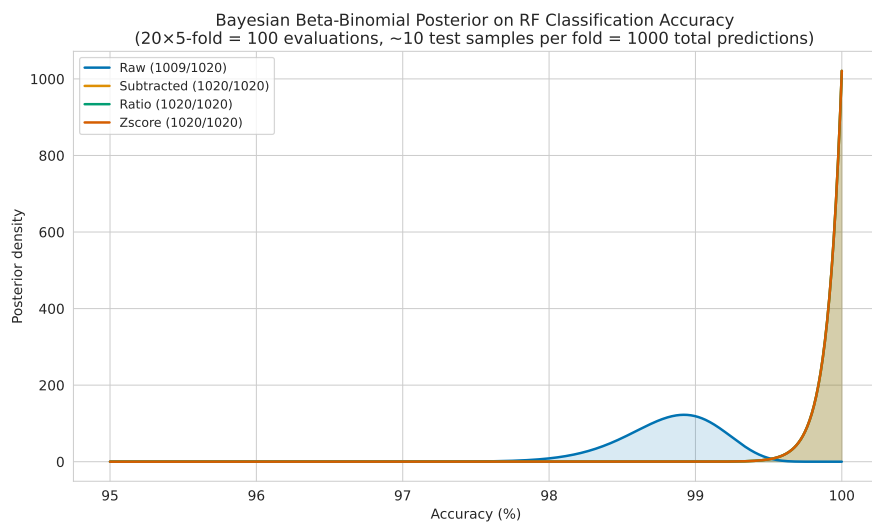


Figure 9. Bayesian Beta-Binomial posterior densities on RF classification accuracy for each isolation method. The raw method's posterior is centered at 98.86% with non-trivial mass below 99%, while the three referenced methods concentrate above 99.6%. The posterior probability that any referenced method exceeds raw is ≥ 0.9997 .

The posterior accuracy difference between any referenced method and raw has median 1.05% with a 95% credible interval of [0.46%, 1.84%], and $P(\text{referenced} > \text{raw}) \geq 0.9997$. This Bayesian framing provides three advantages over the frequentist tests: (1) it produces a properly bounded credible interval despite the ceiling effect in the referenced methods, replacing the degenerate frequentist CI [1.000, 1.000] with the principled [0.9964, 0.99997]; (2) it gives a direct probability statement about the comparison rather than a p -value against a null hypothesis; and (3) it makes explicit that the magnitude of the effect, while small in absolute terms (~ 1 percentage point), is overwhelmingly likely to be a real positive effect rather than noise. One caveat: the Beta-Binomial model treats the 1,020 fold-level predictions as independent Bernoulli trials, but they derive from only 51 runs evaluated ~ 20 times each (raw's 11 errors trace largely to a single borderline run). The effective sample size is therefore closer to 51 than 1,020, so the credible interval and $P(\text{referenced} > \text{raw})$ should be read as optimistic upper bounds on certainty rather than calibrated probabilities; the run-level Cliff's-delta analysis, which localizes the effect to one run, is the more conservative summary.

5.20. Temporal Fingerprints

Figure 10 shows the per-class temporal fingerprints of the most discriminative position-proxy channel (frame_r2.Mx) across the 20 temporal bins, with each toolpath type's individual runs and mean $\pm$ std, in both the raw-active and air-cut-subtracted representations.



Figure 10. Temporal fingerprints of the top-ranked position-proxy channel (frame_r2.Mx mean) across 20 temporal bins, by toolpath type (rows: adaptive, facing, pocketing) and representation (left: raw active; right: air-cut subtracted). Thin lines are individual runs; the bold line and shaded band are the per-type mean $\pm$ one standard deviation. The three types occupy distinct magnetometer ranges—a workspace-position effect (Section 5.7)—and air-cut subtraction (right column) compresses the inter-type offsets while retaining a residual separation.

Figure 11 presents the temporal fingerprint overlay, showing how the three toolpath types produce distinct quasi-static sensor signal profiles across normalized program time.

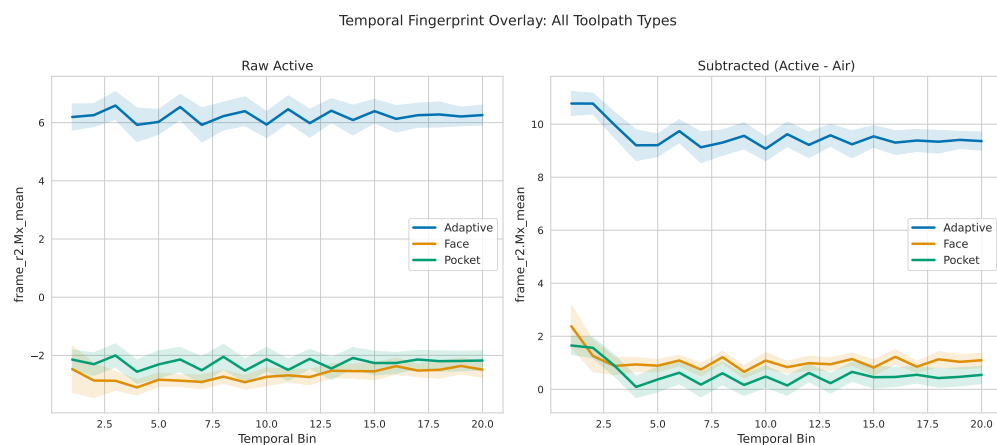


Figure 11. Temporal fingerprint overlay for all three toolpath types. Left: raw active signals. Right: subtracted (air-cut referenced) signals. Shaded regions: ± 1 std. The distinct separation between classes persists after subtraction. Note that these profiles reflect quasi-static motion envelopes rather than cutting vibration dynamics (Section 4.3.3).

Adaptive clearing produces the highest absolute magnetometer values (reflecting its workspace location) with gradual temporal variation, while facing and pocketing occupy a different region with

more temporal structure. After air-cut subtraction (right panel), the inter-class separation is reduced but still clearly present, confirming that some discriminative information survives the referencing process and is not purely position-dependent.

5.21. Generalization Analysis

Table 18 reports leave- K -runs-out CV results.

Table 18. Leave- K -runs-out accuracy (RF). For $K \geq 2$, results are averaged over 100 random combinations.

Method	$K = 1$	$K = 2$	$K = 3$
Raw	100.0 ± 0.0	99.5 ± 5.0	99.0 ± 5.7
Subtracted	100.0 ± 0.0	100.0 ± 0.0	100.0 ± 0.0
Ratio	100.0 ± 0.0	100.0 ± 0.0	100.0 ± 0.0
Z-score	100.0 ± 0.0	100.0 ± 0.0	100.0 ± 0.0

Raw features degrade slightly at $K = 3$ (99.0%), while all referenced methods maintain 100%—consistent with the repeated CV finding that air-cut referencing improves robustness.

5.22. Harder Classification Tasks

To probe the limits of the methodology beyond the easy three-class task, we evaluate several more challenging classification scenarios.

5.22.1. Six-Class Classification (Toolpath $\times$ Condition)

Combining the 55 air-cut and 51 active-cut runs into a six-class problem (3 toolpath types $\times$ 2 cutting conditions), the RF classifier achieves $99.1\% \pm 1.8\%$ accuracy. This demonstrates that sensor features can simultaneously distinguish both the toolpath type and the cutting condition (air vs. active), with near-perfect separability.

5.22.2. Air vs. Active Discrimination

Binary classification of air-cut vs. active-cut runs (ignoring toolpath type) achieves $99.0\% \pm 1.9\%$ accuracy, confirming that material engagement produces a detectable change in sensor signatures across all toolpath types. Per-toolpath binary discrimination (air vs. active within each type) achieves 100% for all three types.

5.22.3. Cross-Condition Transfer

A critical test for the air-cut referencing hypothesis is whether models transfer across cutting conditions. We train an RF classifier on *air-cut data only* (55 runs, 3 toolpath classes) and test on *active-cut data* (51 runs):

Table 19. Cross-condition transfer accuracy. Models trained on one condition are tested on the other without retraining.

Transfer Direction	Accuracy (%)	Train / Test
Air $\rightarrow$ Active	37.3	55 / 51
Active $\rightarrow$ Air	40.0	51 / 55

Both transfer directions achieve near-chance accuracy (~ 37 – 40% vs. 33% random baseline), demonstrating that raw sensor features learned from one cutting condition do not transfer to the other. This result has two important implications: (1) it confirms that the air and active sensor signals are fundamentally different—the machine dynamics (captured by air cuts) and process dynamics

(added by active cuts) produce distinct feature distributions; and (2) it motivates the air-cut referencing methodology, since the process component isolated by subtraction/normalization may be more transferable than the raw signal.

5.22.4. Leave-One-Type-Out: Unknown Toolpath Detection

We train a binary classifier on two toolpath types and examine its behavior when presented with the unseen third type. Since the held-out class was not in the training data, the classifier must assign it to one of the known classes.

Table 20. Leave-one-type-out analysis. An RF classifier trained on two toolpath types is applied to the unseen third type. “Predicted as” shows how the unknown type is classified; “Confidence” reports the RF posterior probability.

Held Out	Predicted As	Mean Conf.	Min Conf.
Adaptive (17)	Pocket: 17/17	0.77 ± 0.05	0.67
Face (15)	Pocket: 15/15	0.79 ± 0.04	0.71
Pocket (19)	Adaptive: 16, Face: 3	0.63 ± 0.06	0.52

When adaptive or face toolpaths are held out, the classifier consistently predicts “pocket” with moderate confidence (0.67–0.79). When pocket is held out, predictions split between adaptive (84%) and face (16%) with lower confidence (min 0.52). The reduced confidence for unknown types suggests that a confidence-thresholding approach could potentially detect novel toolpath types—a direction for future work in anomaly detection.

6. Ablation Study

To understand which components of the sensing and feature extraction pipeline are critical for classification performance, we conduct a comprehensive ablation study. All ablation experiments use the RF classifier with the raw isolation method (the best-performing configuration without PCA).

6.1. Sensor Group Ablation

Table 21 reports classification accuracy when restricting the feature set to individual sensor groups.

Table 21. Classification accuracy by sensor group. “All” uses all 110 sensor channels. Individual Arduino modules use 17 channels each. Electrical uses 8 current sensors.

Sensor Group	Accuracy (%)	Features
All sensors	100.0 ± 0.0	1,540
Arduino (all 6 modules)	100.0 ± 0.0	1,428
y_bed_3	100.0 ± 0.0	238
frame_l2	98.0 ± 4.0	238
y_bed_4	98.0 ± 4.0	238
frame_r2	96.0 ± 4.9	238
spindle2	96.0 ± 4.9	238
frame_l3	94.0 ± 8.0	238
Electrical (current only)	76.5 ± 12.4	112

The y_bed_3 sensor module alone achieves 100% accuracy with only 238 features, demonstrating that a single well-placed IMU on the machine bed is sufficient for perfect toolpath classification. This finding has practical significance: a minimal sensor deployment of one IMU near the workpiece can match the performance of the full 10-sensor array. The electrical current sensors alone achieve only 76.5% accuracy (112 features; corrected from an earlier leaked figure of 96.0%/350 features that had folded IMU channels into the electrical set, Section 5), confirming that motor-current monitoring is

substantially less discriminative than vibration sensing on this task—consistent with the near-constant spindle current under the shallow UHMW-PE cut.

6.2. Feature Statistic Ablation

Table 22 examines how many summary statistics per temporal bin are needed.

Table 22. Classification accuracy as a function of the number of summary statistics computed per temporal bin.

Statistic Set	Accuracy (%)	Features
Mean + RMS (2 stats)	98.0 ± 4.0	440
Mean + RMS + Std (3 stats)	98.0 ± 4.0	660
All 7 statistics	100.0 ± 0.0	1,540

Even with only two statistics (mean and RMS), classification accuracy reaches 98%, indicating that the temporal bin means and RMS values capture most of the discriminative information. The full 7-statistic set adds higher-order statistics (skewness, kurtosis) that contribute the final 2% improvement.

6.3. Alignment Ablation: Temporal Binning vs. Whole-Run Summary

Table 23 compares the temporal binning approach against whole-run summary statistics.

Table 23. Classification accuracy for temporal binning ($B = 20$, aggregated to mean + std across bins) vs. whole-run summary statistics (no temporal structure).

Approach	Accuracy (%)	Features
Temporal binning (20 bins)	100.0 ± 0.0	1,540
Whole-run summary	98.0 ± 4.0	770

Temporal binning outperforms whole-run summary by 2 percentage points. While modest, this confirms that the temporal evolution of sensor statistics carries discriminative information. The temporal approach also enables the fingerprint visualizations (Figure 10) that provide interpretable insight into how toolpath types differ.

6.4. PCA Ablation

Table 24 examines the effect of PCA dimensionality reduction across all three classifiers.

Table 24. Classification accuracy (%) vs. PCA component count, for all three classifiers with raw features. RF shows invariance to component count (all reduce to 88.2%), while SVM and MLP exhibit different sensitivity patterns.

Components	RF	SVM	MLP
No PCA (1,540)	100.0 ± 0.0	90.0 ± 6.3	86.0 ± 10.2
2	90.2 ± 0.4	90.2 ± 0.4	58.9 ± 2.2
3	88.2 ± 4.1	90.2 ± 6.3	74.2 ± 12.2
5	88.2 ± 4.1	94.0 ± 4.9	76.2 ± 10.5
10	88.2 ± 4.1	92.0 ± 7.5	90.0 ± 12.6
15	88.2 ± 4.1	92.0 ± 7.5	92.0 ± 7.5
20	88.2 ± 4.1	92.0 ± 7.5	88.2 ± 4.1
30	88.2 ± 4.1	92.0 ± 7.5	80.5 ± 13.8

The PCA results reveal classifier-dependent behavior. RF accuracy drops to exactly 88.2% for any number of PCA components from 3 to 30, because the leading principal components contain sufficient class-separating variance for the RF decision boundaries, and additional components add only noise

within the PCA-reduced space. In contrast, SVM accuracy *improves* with PCA to 5 components (94.0%), likely because PCA removes irrelevant high-dimensional noise that the RBF kernel is sensitive to. MLP shows non-monotonic behavior, peaking at 15 components (92.0%) before degrading due to overfitting in higher dimensions with only 51 training samples. These results demonstrate that dimensionality reduction interacts strongly with classifier choice, and that the “PCA hurts RF” finding should not be generalized to all classifiers.

PCA misclassification analysis. To explain why RF accuracy is invariant to PCA component count, we examined which individual samples are misclassified across all PCA configurations. The same 5–6 samples are misclassified in *every* PCA setting (3 through 30 components): pocket150025_001, pocket150025_002, pocket150025_012, face150025_001, face150025_010, and (in most settings) pocket150025_006. These runs are consistently confused between the pocketing and facing classes. This confirms that the invariance is not a statistical artifact but reflects a genuine decision boundary: PCA projects the data into a subspace where these specific runs fall near the class boundary, and no amount of additional PCA dimensions resolves the ambiguity. Without PCA, the full 1,540-dimensional feature space provides sufficient redundancy for RF to correctly classify these borderline runs.

6.5. Learning Curve

Figure 12 shows classification accuracy as a function of training set size.

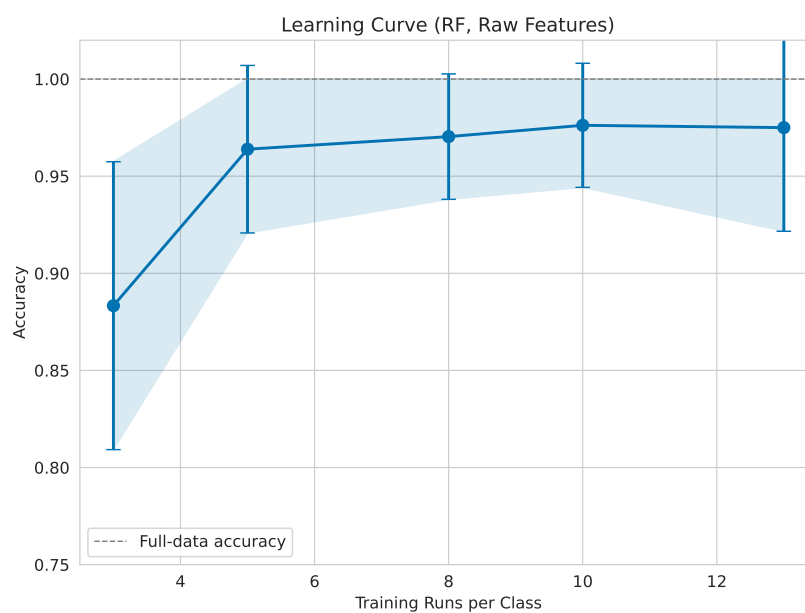


Figure 12. Learning curve: classification accuracy vs. number of training runs per class, using RF with raw features. Results are averaged over 10 random subsampling repeats.

With only 5 training runs per class (15 total), the classifier achieves 96.4% accuracy. Performance plateaus above 8 runs per class at approximately 97%. This steep learning curve demonstrates that the toolpath fingerprints are highly distinctive, requiring minimal training data.

6.6. Sensor Contribution Heatmap

Figure 13 provides a comprehensive view of how each sensor group performs across all four isolation methods.

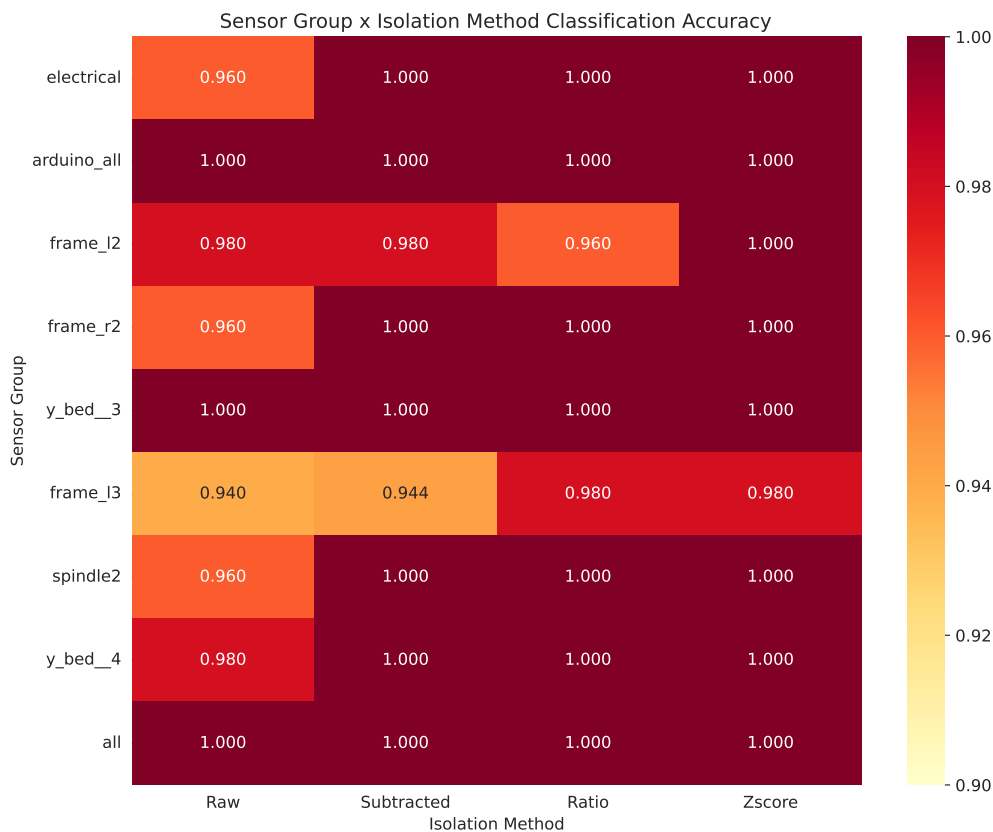


Figure 13. Classification accuracy heatmap by sensor group (rows) and isolation method (columns), using the RF classifier. Darker colors indicate higher accuracy.

The heatmap reveals that sensor performance is relatively consistent across isolation methods, suggesting that the sensor placement (physical position on the machine) is more important than the signal normalization strategy for this classification task.

6.7. Noise Robustness

To assess how sensitive each isolation method is to measurement noise, we add zero-mean Gaussian noise scaled by increasing fractions of the per-feature standard deviation and measure classification accuracy (Figure 14).

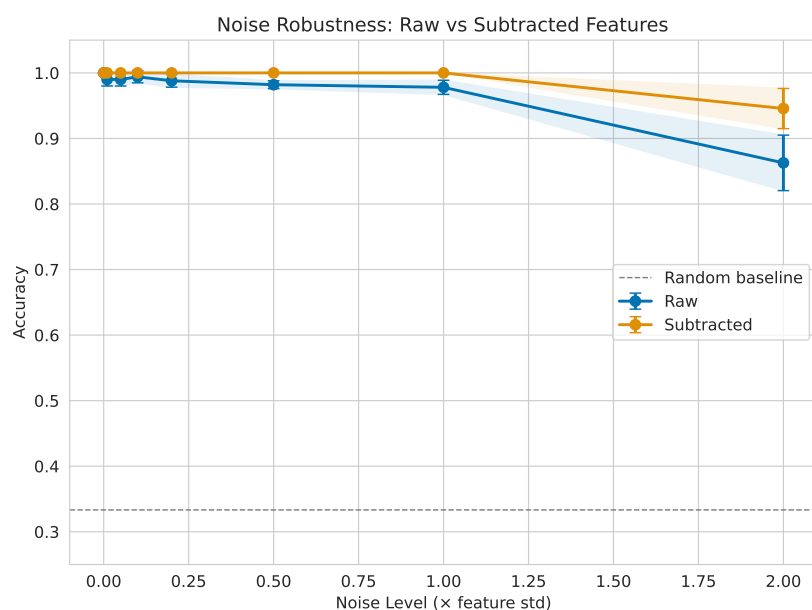


Figure 14. Noise robustness: classification accuracy vs. additive Gaussian noise level (as a fraction of per-feature standard deviation). Subtracted features maintain 100% accuracy up to $1.0\times$ noise, while raw features degrade to 97.8%.

Subtracted features are dramatically more noise-robust than raw features: they maintain 100% accuracy up to $1.0\times$ noise (where the added noise equals the feature's natural standard deviation), while raw features degrade to 97.8% at the same level. At $2.0\times$ noise, subtracted features still achieve 94.6% vs. 86.3% for raw. This result provides the strongest practical argument for air-cut referencing: the referenced features are substantially more resilient to sensor degradation, calibration drift, and environmental interference.

6.8. Temporal Bin Importance

Figure 15 shows the accuracy impact of removing each of the 20 temporal bins.

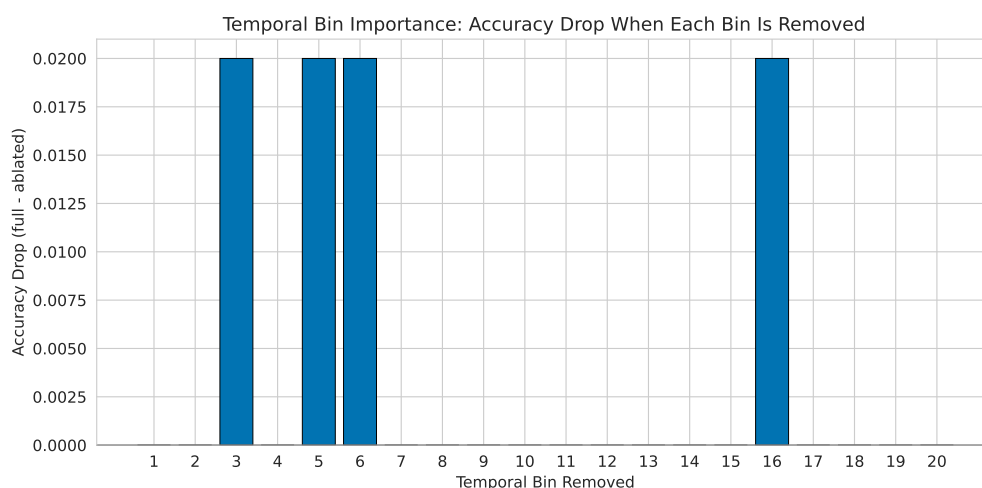


Figure 15. Accuracy drop when each temporal bin is removed (full accuracy minus leave-one-bin-out accuracy). Positive values indicate bins that contribute to classification. Bins 3, 5, 6, and 16 contribute most (each a 2-pp accuracy drop when removed).

Four bins share the largest impact: bins 3, 5, 6, and 16 (1-indexed), each causing a 2 percentage point drop to 98.0% when removed. These correspond to approximately 10–30% of normalized

program time (the approach and early cutting phases) and 75–80% (late cutting phase). The remaining 16 bins can each be removed without any accuracy loss, confirming that the temporal fingerprint’s discriminative power is concentrated in specific program phases rather than distributed uniformly across the execution. This finding has practical implications: a coarser binning strategy focused on these critical phases could achieve comparable accuracy with fewer features.

Figure 16 summarizes the core ablation results in a composite figure.

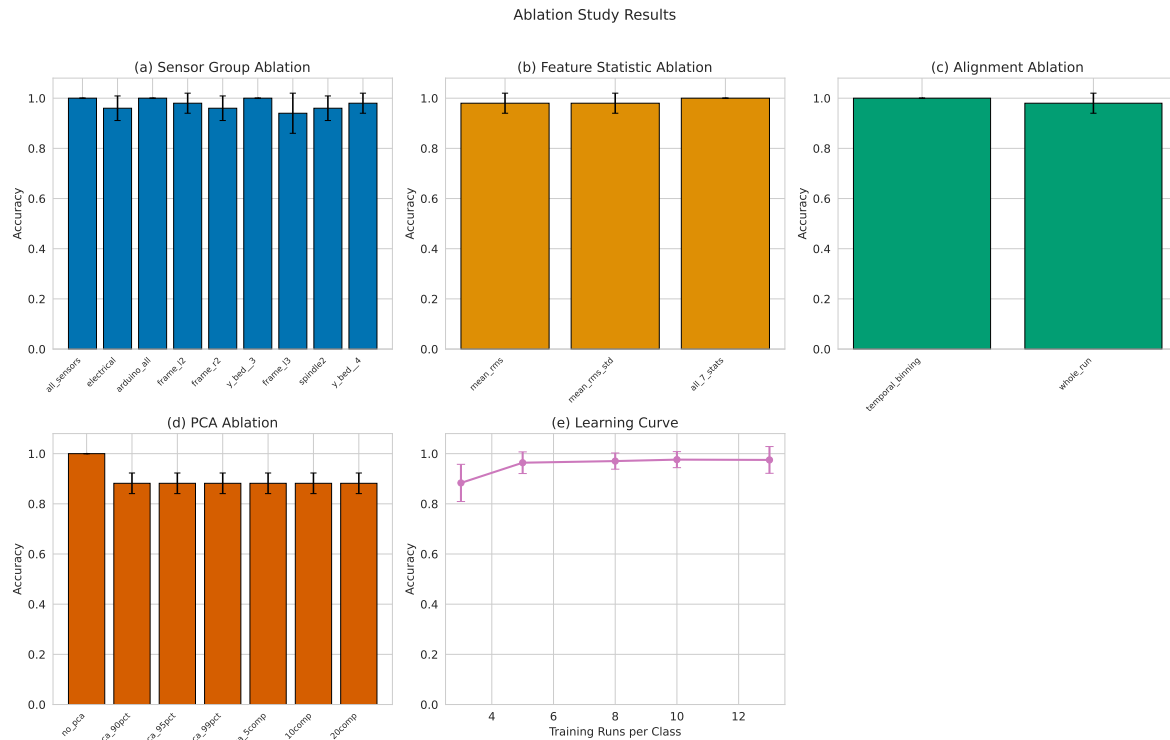


Figure 16. Summary of core ablation studies: (a) sensor group, (b) feature statistics, (c) alignment method, (d) PCA configuration, (e) learning curve.

7. Discussion

7.1. Task Difficulty and the Role of Trivial Baselines

The trivial baseline analysis (Table 3) is the most important methodological result of this study: the three toolpath types are distinguishable from program metadata alone—G-code line range, mean axis position, or even row count. This means that the 100% RF accuracy achieved by our sensor features, while technically correct, does not by itself demonstrate the value of sensor-based classification or air-cut referencing for this specific three-class task.

We report this finding transparently because it establishes the true difficulty floor for this classification problem and prevents over-claiming. The sensor-based approach remains valuable for three reasons: (1) it works without access to program metadata, which may be unavailable or unreliable in deployment scenarios; (2) it provides the temporal fingerprint representation that offers interpretable physical insight; and (3) it establishes a methodology that transfers to harder discrimination tasks where metadata cannot help—demonstrated directly by the damaged-tool detection result (Section 5.8), where the toolpath programs are identical so neither program metadata nor workspace position can separate the classes, yet the sensors reach 98.5% accuracy (balanced 0.99) using process-dynamics channels (accelerometer and acoustic RMS, 74% of importance) rather than the magnetometer position encoder.

7.2. Effectiveness of Air-Cut Referencing

Despite the easy task, three lines of evidence demonstrate a genuine contribution of air-cut referencing:

Statistical significance (a consistency check, not the main evidence). On the full feature set the air-cut effect is small and ceiling-bound: the Friedman test ($\chi^2 = 33.0$, $p < 10^{-6}$, $p_{\text{Bonf}} = 0.011$) is significant and the bootstrap CIs are non-overlapping (raw [0.982, 0.994] vs. referenced [1.000, 1.000]), but the gap is only ~ 1 pp at a 100% ceiling, Cliff's $\delta = 0.022$ is negligible, and the significance traces to a single recurrently-misclassified run. Cohen's d is not meaningful here (the referenced methods have zero variance at the ceiling), so we do not interpret it. We therefore treat the full-feature tests as a consistency check and rest the case for air-cut referencing on the confound-free experiment (raw RF 96.7%, min 81.8%, vs. subtracted/ratio 100%) together with the noise-resilience and data-efficiency results below.

Robustness. Across 100 repeated CV evaluations, raw RF features occasionally misclassify (minimum fold accuracy 90%), while referenced methods maintain perfect accuracy. Per-sample analysis traces this to a single borderline run (pocket150025_001) that raw RF misclassifies 55% of the time but all referenced methods classify correctly 100% of the time.

Noise resilience. The noise robustness analysis (Figure 14) provides the strongest practical argument: subtracted features maintain 100% accuracy up to $1.0\times$ additive noise (equal to the feature's natural standard deviation), while raw features degrade to 97.8%. At $2.0\times$ noise, the gap widens to 94.6% vs. 86.3%. This suggests that air-cut referencing would be particularly valuable in production environments with higher sensor noise, calibration drift, or environmental interference.

The two-way ANOVA reveals that classifier choice ($\eta^2 = 0.17$) explains approximately $4\times$ more variance than isolation method ($\eta^2 = 0.04$), indicating that for this task, selecting the right classifier matters more than the referencing strategy.

For SVM and MLP classifiers, air-cut referencing provides 4–8 percentage point improvements in mean accuracy and substantially reduces variance, suggesting it removes machine-specific baseline variation that these classifiers are sensitive to.

7.3. Temporal Structure in Toolpath Fingerprints

The temporal binning approach outperforms whole-run summary by 2 percentage points (100% vs. 98%), confirming that temporal structure carries discriminative information. The temporal fingerprint visualizations (Figure 10) reveal characteristic patterns: adaptive clearing produces gradual, continuous signal evolution; facing generates periodic oscillations corresponding to linear passes; and pocketing shows distinct phase transitions from contour passes.

Importantly, the use of different G-code programs for air and active conditions necessitated temporal binning rather than point-by-point G-code-aligned subtraction. Despite this constraint, the temporal approach captures meaningful phase alignment (approach, cutting, retract) through normalized time binning. Future data collection with identical programs would enable a comparison of these approaches.

7.4. What the Sensors Are Actually Detecting

The physical interpretation analysis (Section 5.7) reveals a critical finding: the dominant discriminative features are magnetometer channels that primarily encode *workspace position* (proximity to motors and ferromagnetic structure) rather than *cutting dynamics* (forces, vibrations, torque). This is the sensor-domain equivalent of the trivial position baseline—the classifier can distinguish toolpath types largely because the three programs move through different regions of the machine's workspace.

This observation has important implications:

- **For this study:** the high classification accuracy is partly attributable to program geometry rather than process physics. Future work testing on toolpath variants that traverse the same workspace region would isolate the cutting-dynamics contribution.
- **For sensor selection:** magnetometers on CNC machines are highly sensitive to position due to motor fields, making them effective for operation identification but poor for process-independent fingerprinting. Accelerometers and current sensors, though less discriminative in this study, capture signals more directly related to cutting mechanics and would likely perform better on within-type discrimination tasks.
- **For air-cut referencing:** the subtracted (active – air) features partially remove position-dependent offsets since the air-cut reference captures the same workspace trajectory. This may explain why the noise robustness analysis (Figure 14) shows referenced features outperforming raw features—the subtraction removes position-correlated noise components.

Spindle current, despite being the most direct proxy for cutting torque, shows almost no variation across toolpath types (5.245–5.257 A). This is consistent with the shallow depth of cut (0.025") producing minimal load relative to the spindle's no-load draw at 12,000 RPM. In industrial machines with heavier cuts, spindle current would likely become a more discriminative channel.

7.5. Sensor Deployment Guidelines

The ablation study provides actionable guidance for deploying toolpath fingerprinting systems:

- A single bed-mounted IMU (y_bed__3) achieves 100% accuracy—the optimal single-sensor placement due to its proximity to both the workpiece and the Y-axis motor.
- Any single IMU module achieves $\geq 94\%$ accuracy regardless of placement, demonstrating robustness.
- Electrical current sensors alone achieve only 76.5% (the weakest sensor group; an earlier 96.0% figure was inflated by the spindle2 feature leak, Section 5), so IMU-based sensing is clearly preferable where it can be installed.
- Two statistics (mean + RMS) per temporal bin achieve 98%; the full 7-statistic set reaches 100%.
- PCA hurts RF (88.2% vs. 100%) but helps SVM (94% at 5 components) and MLP—dimensionality reduction must be matched to the classifier.

7.6. Temporal Drift and Run-Order Effects

Run-order analysis reveals statistically significant temporal drift in several sensor channels, particularly for the adaptive clearing class: temperature correlates strongly with run order (Spearman $r = 0.98$, $p < 0.001$), and the bed accelerometer Az shows similar drift ($r = 0.97$). This indicates that the machine warms up over the course of a recording session, producing systematic feature changes correlated with run number rather than toolpath type. Since runs within each toolpath class were collected sequentially, this temperature drift is partially confounded with within-class ordering but does not create between-class confounds (all three types show warming trends).

7.7. Error Analysis: Borderline Run pocket150025_001

Per-sample analysis identified pocket150025_001 as the single run that raw RF misclassifies in 55% of repeated CV evaluations. Feature-level analysis reveals that this run deviates from the pocketing class centroid by $> 3.9\sigma$ on 15 features—primarily color sensor variability (ColorR, ColorG, ColorB, ColorA) and frame_l3 gyroscope statistics. On all 15 deviating features, the sample's value is closer to the facing class centroid than to the pocketing centroid. The color sensor dominance suggests an ambient lighting anomaly during this specific run (e.g., a shadow change). Air-cut referencing resolves this misclassification by normalizing out the session-specific color baseline, demonstrating its value for handling run-level environmental variability.

7.8. Causal Structure of the Classification Problem

The empirical findings of this study—trivial baselines achieving 100%, magnetometer dominance via position encoding, and the confound-free pipeline still benefiting from air-cut referencing—are coherently explained by the causal structure shown in Figure 17. The G-code program (an exogenous variable corresponding to the toolpath type) determines both the sequence of motion commands and the workspace region traversed. Workspace region in turn determines magnetometer readings (via proximity to stepper motors and ferromagnetic structure) and motor current dynamics (via different acceleration profiles). Cutting forces are an additional contribution from material removal, which under our shallow-DOC UHMW-PE conditions is small relative to machine-specific signals.

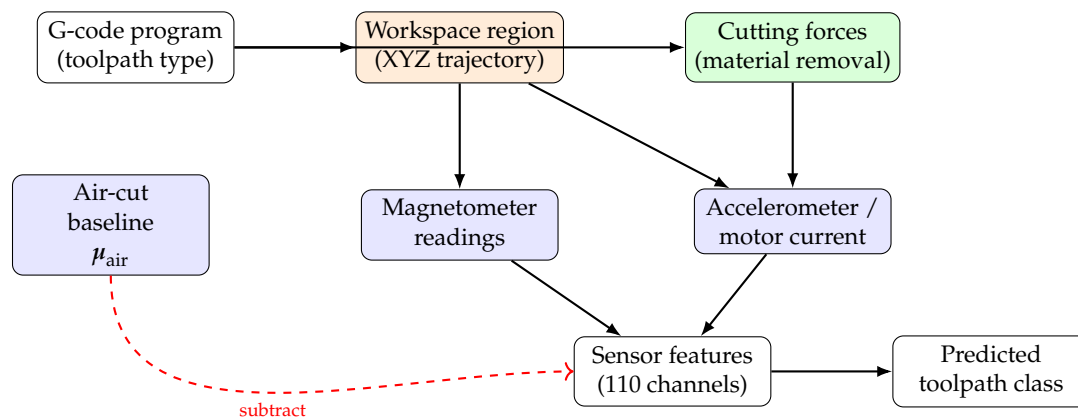


Figure 17. Causal structure of the toolpath classification task. Solid arrows represent causal dependencies in the data-generating process. The G-code program is the root cause of both workspace region (a confound) and cutting forces (the process-relevant signal). Magnetometer features are downstream of workspace region only, making them position encoders. Accelerometer and motor-current features are downstream of both workspace region and cutting forces. The dashed red arrow represents the air-cut referencing intervention: subtracting the air-cut baseline from active features removes the workspace-dependent component, leaving the cutting-force component. Because the air-cut and active programs in our dataset traverse *similar but not identical* workspace regions, this subtraction is partial (Section 5). Note that under the shallow-DOC UHMW-PE conditions of this study, the cutting-forces → motion-features arrow carries a small signal (~1–2 N at the tool); on metallic workpieces with realistic cutting forces (50–500 N), this arrow would be substantially stronger, shifting the relative balance of position-encoded versus force-encoded discrimination.

This structural model explains four otherwise puzzling observations. First, magnetometer features dominate feature importance because they are directly downstream of the workspace-region confound, which carries nearly all the class signal. Second, removing magnetometer channels does not destroy classification accuracy because accelerometer and motor-current features are also downstream of workspace region (via trajectory-dependent acceleration patterns). Third, air-cut referencing helps even on confound-free features because it removes workspace-dependent components from the accelerometer and current channels as well, isolating their force-relevant residuals. Fourth, cross-condition transfer fails (37% accuracy) because the air-cut and active-cut conditions used different G-code programs and thus produced different workspace regions, so the position-encoded features do not transfer; the small force-relevant signal would transfer but is dominated by the position component in the raw representation. Together these explain why the paper’s headline result (100% accuracy) is not the contribution—the contribution is identifying the causal structure that makes such results possible and providing a methodology for separating the position-encoded path from the force-relevant path.

7.9. Limitations and Future Work

No independent test set. Beyond the single internal held-out split (Section 5: 12 of the 51 runs, evaluated once), all evaluations rely on cross-validation (5-fold stratified, 20 repeats) on the same 51 active-cutting runs. That internal split shares the machine, session, and CAM programs of the training data, so it guards against fold-reuse leakage but is *not* an independent test of generalization; and with 100 CV evaluations every sample appears in many train/test splits. An *independent* test set from a future collection round (a new session, and ideally a second machine) would provide stronger evidence of generalization. Within the existing data, the leave- K -runs-out analysis ($K \leq 3$) is the closest approximation.

Single machine. All data is collected from one Bantam Tools Explorer CNC mill. The central hypothesis—that air-cut referencing produces machine-independent fingerprints—cannot be fully validated without cross-machine experiments. The strong generalization results (leave- K -out accuracy) suggest robustness to within-machine variation but do not address between-machine variation.

Different CAM programs. The air-cut and active-cutting data used different CAM-generated G-code programs, preventing point-by-point signal subtraction. The temporal binning approach compensates for this, but future data collection should use identical G-code programs for both conditions to enable direct comparison.

Limited toolpath diversity. Three toolpath types with fixed cutting parameters (0.150" stepover, 0.025" depth) provide a clean classification task but do not test discrimination between subtle variations within a toolpath type.

Workpiece material and cutting forces. The UHMW-PE workpiece produces extremely low cutting forces ($\sim 1\text{--}2$ N for a 6.35 mm endmill at 0.635 mm DOC) compared to metals. This means the “process-specific signal” isolated by air-cut referencing is dominated by trajectory differences and quasi-static motion dynamics rather than cutting force signatures. On metallic workpieces with higher cutting forces, the sensor signals would contain a larger cutting-force component, potentially making the accelerometer and current channels more discriminative relative to the magnetometer position proxy.

Low sampling rate and decimation. The 4 Hz effective sampling rate (decimated from 119 Hz IMU and 1 kHz DAQ without explicit anti-aliasing) captures only quasi-static machine motion, not cutting vibration. Higher-rate acquisition retaining the native DAQ bandwidth would enable frequency-domain features (e.g., power spectral density at cutting-relevant frequencies) that are unavailable in this study.

Small dataset. With 51 active-cutting runs and ≤ 19 runs per class, the dataset is adequate for the current three-class problem but may be insufficient for more complex tasks or deep learning approaches.

Near-perfect accuracy. The 100% RF accuracy across all methods raises concerns about task difficulty. The three toolpath types are structurally quite different (continuous adaptive, back-and-forth facing, concentric pocketing), making them inherently easy to distinguish. Testing on more similar toolpath variants would better evaluate the methods’ discriminative power.

8. Conclusions

This paper introduced air-cut referencing with temporal binning as a methodology for extracting toolpath fingerprints from CNC sensor data. We evaluated four signal isolation methods across three classifiers for three-class toolpath classification (adaptive clearing, facing, pocketing) on a dataset of 51 active-cutting and 55 air-cutting runs from a desktop CNC mill.

The key findings are:

1. **Task difficulty context.** Trivial baseline analysis reveals that the three toolpath types are separable from program metadata alone (G-code line range: 100%, mean position: 98%). This contextualizes

the high sensor-based accuracy and underscores that the primary contribution is methodological rather than accuracy-driven.

2. **Air-cut referencing provides genuine, confound-independent benefit.** When all position-confounded channels (magnetometers, color, pressure, proximity) are excluded (784 features), raw RF drops to 96.7% with a minimum fold accuracy of 81.8%, while the subtracted and ratio methods maintain **perfect 100% accuracy across all 100 repeated CV evaluations** and z-score remains near-perfect (99.0%, minimum 90%). This demonstrates that the referencing benefit is not an artifact of the magnetometer position-encoding confound. The Friedman test ($p < 10^{-6}$) confirms statistical significance, and noise robustness analysis shows referenced features maintain accuracy up to $1.0 \times$ noise.
3. **Process dynamics are recoverable once position is controlled.** On a damaged-tool detection task using the same G-code programs (so workspace position is held constant), Random Forest reaches 98.5% accuracy (balanced 0.99, damage-class $F_1 = 0.97$), with discriminative importance shifting from the magnetometer (0.1%) to the accelerometer and acoustic-RMS channels (74% combined). This is the most process-relevant result and a direct tool-condition-monitoring application; air-cut referencing is neutral here, consistent with the causal model (referencing helps only when a position confound is present).
4. **Temporal structure is informative.** Temporal binning (20 bins) outperforms whole-run summary statistics (100% vs. 98%), and produces interpretable fingerprint visualizations that reveal toolpath-specific sensor dynamics.
5. **Minimal sensor requirements.** A single bed-mounted IMU achieves 100% accuracy with 238 features. Only 5 training runs per class suffice for 96.4% accuracy.
6. **Competitive with modern methods.** ROCKET-style random convolutional features achieve comparable accuracy (98–100%), confirming the task’s accessibility while validating that our interpretable approach does not sacrifice performance.

A notable limitation is that the air-cut and active-cutting data used different CAM-generated G-code programs, preventing direct G-code-aligned signal subtraction. The temporal binning approach compensates for this, but future data collection should use identical programs.

Future work will focus on: (1) within-toolpath-type parameter discrimination (e.g., different stepovers), where trivial baselines cannot help and air-cut referencing should provide greater benefit; (2) cross-machine validation to test transferability of referenced fingerprints; and (3) higher-rate data acquisition to enable frequency-domain features.

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1. Jardine, A.K.S.; Lin, D.; Banjevic, D. A review on machinery diagnostics and prognostics implementing condition-based maintenance. *Mechanical Systems and Signal Processing* **2006**, *20*, 1483–1510. doi:10.1016/j.ymssp.2005.09.012.

2. Li, X.; Zhang, W.; Ding, Q.; Sun, J.Q. Intelligent rotating machinery fault diagnosis based on deep learning using data augmentation. *Journal of Intelligent Manufacturing* **2020**, *31*, 433–452. doi:10.1007/s10845-018-1456-1.
3. Teti, R.; Jemielniak, K.; O'Donnell, G.; Dornfeld, D. Advanced monitoring of machining operations. *CIRP Annals* **2010**, *59*, 717–739. doi:10.1016/j.cirp.2010.05.010.
4. Byrne, G.; Dornfeld, D.; Inasaki, I.; Ketteler, G.; König, W.; Teti, R. Tool condition monitoring (TCM)—the status of research and industrial application. *CIRP Annals* **1995**, *44*, 541–567. doi:10.1016/S0007-8506(07)62503-4.
5. Nath, C. Integrated tool condition monitoring systems and their applications: a comprehensive review. *Procedia Manufacturing* **2020**, *48*, 953–962. doi:10.1016/j.promfg.2020.05.135.
6. Kuntoğlu, M.; Aslan, A.; Pimenov, D.Y.; Usca, Ü.A.; Salur, E.; Gupta, M.K.; Mikolajczyk, T.; Giasin, K.; Kapłonek, W.; Sharma, V.S. A review of indirect tool condition monitoring systems and decision-making methods in turning: critical analysis and trends. *Sensors* **2021**, *21*, 108. doi:10.3390/s21010108.
7. Möhring, H.C.; Bertram, O. Integrated autonomous monitoring of ball screw drives. *CIRP Annals* **2012**, *61*, 355–358. doi:10.1016/j.cirp.2012.03.138.
8. Cai, Y.; Starly, B.; Cohen, P.; Lee, Y.S. Sensor data and information fusion to construct digital-twins virtual machine tools for cyber-physical manufacturing. *Procedia Manufacturing* **2017**, *10*, 1031–1042. doi:10.1016/j.promfg.2017.07.094.
9. Sick, B. On-line and indirect tool wear monitoring in turning with artificial neural networks: a review of more than a decade of research. *Mechanical Systems and Signal Processing* **2002**, *16*, 487–546. doi:10.1006/mssp.2001.1460.
10. Zhou, Y.; Xue, W. Review of tool condition monitoring methods in milling processes. *The International Journal of Advanced Manufacturing Technology* **2023**, *125*, 4965–4995. doi:10.1007/s00170-023-11054-2.
11. Zhao, R.; Yan, R.; Chen, Z.; Mao, K.; Wang, P.; Gao, R.X. Deep learning and its applications to machine health monitoring. *Mechanical Systems and Signal Processing* **2019**, *115*, 213–237. doi:10.1016/j.ymssp.2018.05.050.
12. Dimla, D.E. Sensor signals for tool-wear monitoring in metal cutting operations—a review of methods. *International Journal of Machine Tools and Manufacture* **2000**, *40*, 1073–1098. doi:10.1016/S0890-6955(99)00122-4.
13. Kuo, R.J. Multi-sensor integration for on-line tool wear estimation through artificial neural networks and fuzzy neural network. *Engineering Applications of Artificial Intelligence* **2000**, *13*, 249–261. doi:10.1016/S0952-1976(00)00008-7.
14. Al-Badour, F.; Sunar, M.; Cheded, L. Vibration analysis of rotating machinery using time–frequency analysis and wavelet techniques. *Mechanical Systems and Signal Processing* **2011**, *25*, 2083–2101. doi:10.1016/j.ymssp.2011.01.017.
15. Rmili, W.; Ouahabi, A.; Serra, R.; Leroy, R. An automatic system based on vibratory analysis for cutting tool wear monitoring. *Measurement* **2016**, *77*, 117–123. doi:10.1016/j.measurement.2015.09.010.
16. Breiman, L. Random Forests. *Machine Learning* **2001**, *45*, 5–32. doi:10.1023/A:1010933404324.
17. Widodo, A.; Yang, B.S. Support vector machine in machine condition monitoring and fault diagnosis. *Mechanical Systems and Signal Processing* **2007**, *21*, 2560–2574. doi:10.1016/j.ymssp.2006.12.007.
18. Sun, J.; Rahman, M.; Wong, Y.S.; Hong, G.S. Multiclassification of tool wear with support vector machine by manufacturing loss consideration. *International Journal of Machine Tools and Manufacture* **2004**, *44*, 1179–1187. doi:10.1016/j.ijmachtools.2004.04.003.
19. Bustillo, A.; Pimenov, D.Y.; Matuszewski, M.; Mikolajczyk, T. Using artificial intelligence models for the prediction of surface wear based on surface isotropy levels. *Robotics and Computer-Integrated Manufacturing* **2022**, *73*, 102238. doi:10.1016/j.rcim.2021.102238.
20. Wang, J.; Ma, Y.; Zhang, L.; Gao, R.X.; Wu, D. Deep learning for smart manufacturing: methods and applications. *Journal of Manufacturing Systems* **2018**, *48*, 144–156. doi:10.1016/j.jmsy.2018.01.003.
21. Cao, X.; Chen, B.; Yao, B.; Zhuang, S. An overview of deep learning methods for machining process monitoring. *Engineering Applications of Artificial Intelligence* **2024**, *133*, 108274. doi:10.1016/j.engappai.2024.108274.
22. Bergs, T.; Holst, C.; Guber, P.; Herrmann, G. Digital image processing with deep learning for automated cutting tool wear detection. *Procedia Manufacturing* **2020**, *48*, 895–901. doi:10.1016/j.promfg.2020.05.127.

23. Dempster, A.; Petitjean, F.; Webb, G.I. ROCKET: Exceptionally fast and accurate time series classification using random convolutional kernels. *Data Mining and Knowledge Discovery* **2020**, *34*, 1454–1495. doi:10.1007/s10618-020-00701-z.
24. Fawaz, H.I.; Lucas, B.; Forestier, G.; Pelletier, C.; Schmidt, D.F.; Weber, J.; Webb, G.I.; Idoumghar, L.; Muller, P.A.; Petitjean, F. InceptionTime: Finding AlexNet for time series classification. *Data Mining and Knowledge Discovery* **2020**, *34*, 1936–1962. doi:10.1007/s10618-020-00710-y.
25. Pan, S.J.; Yang, Q. A survey on transfer learning. *IEEE Transactions on Knowledge and Data Engineering* **2010**, *22*, 1345–1359. doi:10.1109/TKDE.2009.191.
26. Jiao, J.; Zhao, M.; Lin, J.; Ding, C. A comprehensive review on convolutional neural network in machine fault diagnosis. *Neurocomputing* **2020**, *417*, 36–63. doi:10.1016/j.neucom.2020.07.088.
27. International Organization for Standardization. ISO 230-1:2012 Test Code for Machine Tools—Part 1: Geometric Accuracy of Machines Operating under No-Load or Quasi-Static Conditions. Standard, 2012.
28. Altintas, Y. *Manufacturing Automation: Metal Cutting Mechanics, Machine Tool Vibrations, and CNC Design*, 2nd ed.; Cambridge University Press, 2012.
29. Monostori, L.; Kádár, B.; Bauernhansl, T.; Kondoh, S.; Kumara, S.; Reinhart, G.; Sauer, O.; Schuh, G.; Sihn, W.; Ueda, K. Cyber-physical systems in manufacturing. *CIRP Annals* **2016**, *65*, 621–641. doi:10.1016/j.cirp.2016.06.005.
30. Sturm, L.D.; Williams, C.B.; Camelio, J.A.; White, J.; Parker, R. Cyber-physical vulnerabilities in additive manufacturing systems: a case study attack on the .STL file with human subjects. *Journal of Manufacturing Systems* **2017**, *44*, 154–164. doi:10.1016/j.jmsy.2017.05.007.
31. Zeltmann, S.E.; Gupta, N.; Tsoutsos, N.G.; Maniatakis, M.; Rajendran, J.; Karri, R. Manufacturing and security challenges in 3D printing. *JOM* **2016**, *68*, 1872–1881. doi:10.1007/s11837-016-1937-7.
32. Yampolskiy, M.; King, W.E.; Gatlin, J.; Belikovetsky, S.; Brown, A.; Skjellum, A.; Elovici, Y. Security of additive manufacturing: attack taxonomy and survey. *Additive Manufacturing* **2018**, *21*, 431–457. doi:10.1016/j.addma.2018.03.015.
33. Belikovetsky, S.; Yampolskiy, M.; Toh, J.; Gatlin, J.; Elovici, Y. dr0wned—Cyber-Physical Attack with Additive Manufacturing. Proceedings of the 11th USENIX Workshop on Offensive Technologies (WOOT 17). USENIX Association, 2017.